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Paleomagnetic Data From the Philippine Sea Plate and Their Tectonic Significance

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New paleomagnetic data from Guam and Saipan have been obtained and are used, in conjunction with previously published results from around the Philippine Sea plate, to interpret the origin and history of the Philippine Sea plate. A lower Miocene direction (I=3.5°, D=54.5°, a95=17.4°, k=20.3) and a middle Oligocene direction (I=5.3°, D=68.4°, a95=12.8°, k=52.4) have been obtained from the Palau Islands. Only one reliable result, a middle Oligocene direction (I=15.1°, D=66.1°, a95=11.1°, k=20.1), has been obtained from Guam. Many characteristic remanent magnetic directions have been obtained from the lower Eocene Facpi Formation; however, the results are too scattered (k=4.5) to be considered reliable. Two reliable directions have been obtained from Saipan: a middle Miocene result (I=30.7°, D=28.1°, a95=8.4°, k=64.6), and a lower Eocene result (I=-12.0°, D=43.0°, a95=12.5°, k=16.0). Two lower Eocene results from Chichijima (I=9.5°, D=105.3°, a95=12.6°, k=6.1) and Anijima (I=-0.8°, D=91.7°, a95=16.9°, k=4.9) are the reliable results from the Bonin islands. There are two end-member interpretations for these data: (1) small scale local rotation of blocks along the plate margin, or (2) rotation of the Philippine Sea plate as a whole. To distinguish between these interpretations, we reconstructed the East Philippine Sea province back to its prerifting configuration. The rotation caused by deformation can be estimated by comparing the paleomagnetism of Guam, Saipan, and the Bonin islands to Palau. Prior to reconstruction, the older declination values are scattered. Closing the marginal basins reduces the declination scatter, and all of the islands show increasing clockwise deflection with time. This suggests that the Philippine Sea plate has rotated up to 80° clockwise and moved northward around 20° since the Eocene, and deformational rotation is a second-order feature. The finite pole of rotation of the Philippine Sea plate relative to Eurasia must be somewhere near the southeast corner of the plate, because this is the only location that is compatible with both the change in orientation and paleolatitude for the plate. These data cannot distinguish between a backarc origin or trapped crust origin for the West Philippine Sea province. The existing models require major modification to be consistent with the paleomagnetic data.

INTRODUCTION

The Philippine Sea plate is one of the least understood lithospheric plates on Earth, despite several decades of intense ship-borne geologic and geophysical studies. The plate contains some of the best examples of island arcs, forearcs, and back arc basins in the world, and thus much of our current thinking about the development of these features is influenced by studies of the Philippine Sea plate [*Karig*, 1971]. The motion of the plate also plays a major role in the tectonics of eastern Asia, forming much of the eastern boundary. Sinistral shear in the Philippine archipelago can be related to oblique subduction of the Philippine Sea plate. However, the motion of the plate is difficult to determine because it is almost exclusively surrounded by subduction zones. Understanding the origin and motion of the Philippine Sea plate has broad implications to the understanding of island arcs, backarc basins, and the development of the southeast Asian margin.

The plate consists of two distinct tectonic provinces, the west Philippine Sea province (WPSP) and the east Philippine Sea province (EPSP). The Palau-Kyushu ridge marks the boundary between the dominantly NW-SE trending features of the WPSP and the N-S trending features of the EPSP (Figure 1). This sharp break in structural trend has inspired many questions and speculations about the origin and history of the Philippine Sea plate.

The WPSP comprises a large open basin (West Philippine basin), a remnant spreading center (Central Basin ridge), and several remnant arcs (Oki-Daito and Daito ridges). Identified magnetic anomalies in the West Philippine basin indicate that the age of the basin ranges from Cretaceous to Eocene, however the exact age of spreading varies between authors (compare *Louden* [1976], *Shih* [1980*a*], and *Hilde and Lee* [1984]]. Several contrasting models have been proposed for the origin of the WPSP, probably the more enigmatic of the two provinces. *Uyeda and Ben-Avraham* [1972], *Hilde et al.* [1977], *Uyeda and McCabe* [1983], and *Hilde and Lee* [1984] suggest that the WPSP is a trapped piece of old Pacific plate oceanic crust, isolated from the Pacific/Kula spreading system when the Pacific plate motion changed around 42 Ma [*Clague and Jarrard*, 1973]. The trapping was caused by the initiation of west-dipping subduction (proto-Mariana-arc) along a former north-south transform fault. In contrast to this, *Karig* [1971], *Lewis et al.* [1982], and *Seno and Maruyama* [1984] suggest that the WPSP formed as a back arc basin. More recently *Seno* [1988] suggests that the WPSP is a unique basin in that it has aspects of both a back arc and an open ocean basin. Certainly, the nature and origin of the WPSP is unclear.

Although the mechanism which produced the sharp contrast in structural trends between the WPSP and EPSP is uncertain, the evolution of the EPSP seems to be better understood than the adjacent WPSP. *Karig* [1971, 1975] first proposed that the EPSP is a series of remnant arcs and back arc basins, formed behind the west-dipping Mariana subduction system. The back arc basins include the Parece-Vela and Shikoku basins, and the currently opening Mariana trough. Magnetic anomalies in the Parece-Vela basin indicate that it opened between 30 and 17 Ma [*Mrozowski and Hayes*, 1979], while the Shikoku basin opened slightly later between 24 and 15 Ma [*Watts and Weissel*, 1975; *Kobayashi and Isezaki*, 1976; *Shih*, 1980b]. The Mariana trough began opening around 5 Ma and continues to open today [*Karig et al.*, 1978; *Bibee et al.*, 1980]. Although variations on the original model have been proposed [*Scott 1983*; *Uyeda and McCabe*, 1983; and others], the concept of arc splitting and back arc spreading is central to

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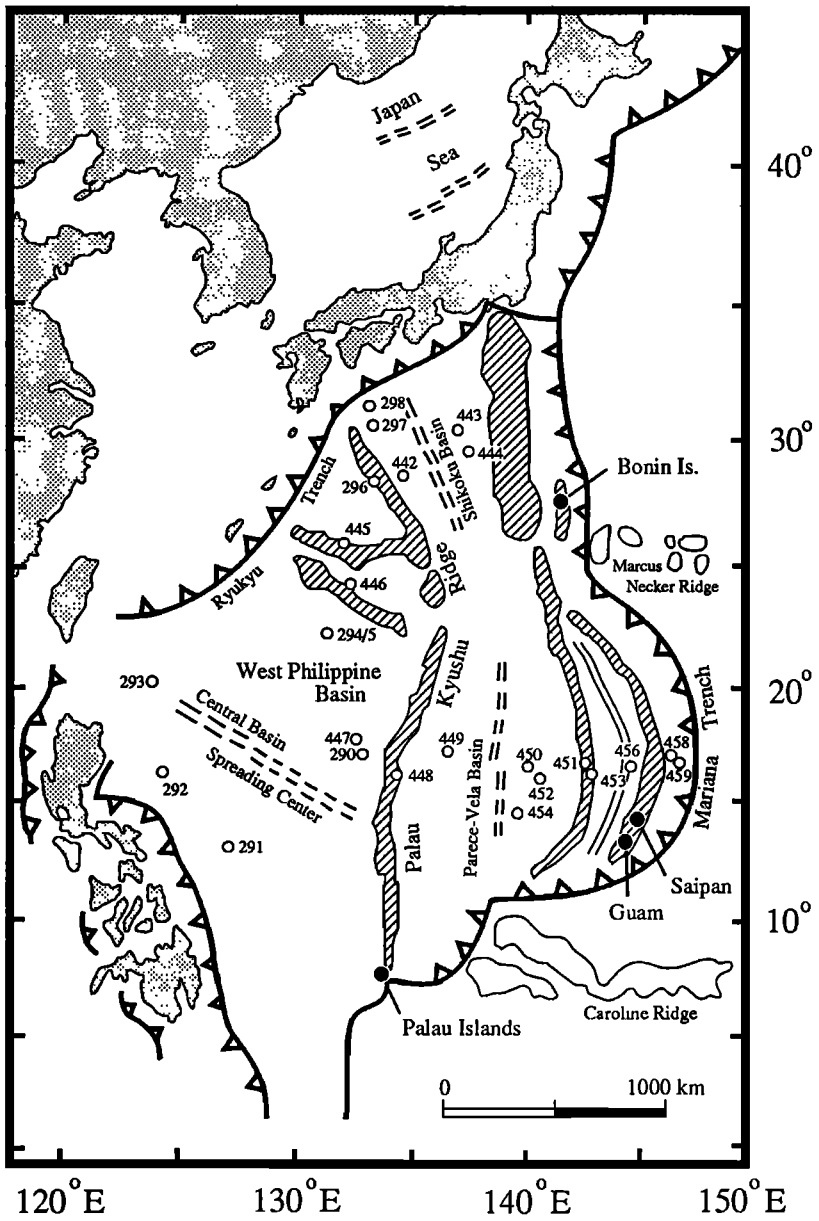


Fig. 1. Major features of the Philippine Sea plate.

most models. These basins have strongly influenced ideas on oceanic arc and backarc basin formation.

Paleomagnetism affords a possible record of the past motion of portions of the Philippine Sea plate and hence its tectonic development. Many paleomagnetic studies have been undertaken on the rocks exposed around the plate. Paleomagnetic data which suggest clockwise rotations along the Mariana and Bonin arcs were taken as evidence that the collision of a buoyant feature, such as an oceanic plateau, with an arc could cause deformational rotation of the arc [*Larson et al.*, 1975; *Vogt et al.*, 1976; *McCabe and Uyeda*, 1983; *McCabe*, 1984]. *Keating and Helsley* [1985] used the same data set to suggest that there has been continuous clockwise rotation of the Philippine Sea plate since the Eocene. *Haston et al.* [1988] used paleomagnetic data from the Palau Islands to postulate that the Philippine Sea plate has rotated clockwise 60°-70° since the middle Oligocene. Inclination studies from Deep Sea Drilling Project (DSDP) drill cores suggest that the plate has moved northward 20° since the Eocene [*Louden*, 1977; *Kinoshita*, 1980; *Bleil*, 1981, and others]. However, the scarcity of subaerial exposures, the paucity of reliable data, and the location of the sampling sites on the plate margin make it difficult to rigorously test these ideas.

This paper presents new paleomagnetic data from around the Philippine Sea plate and synthesizes all of the previously published work, in an effort to determine the past motion of the plate and deformation of the plate margin.

PALEOMAGNETIC METHODS

Paleomagnetic samples were collected on Guam, Saipan, and the Palau Islands. A gasoline-powered drill was used to take 2.5-cm-diameter cores, which were oriented in the field to ±2° by magnetic compass. In this study a site is defined as a set of samples taken from a single cooling unit or sedimentary horizon. Structural attitudes were taken from sedimentary contacts when possible. In units where structural attitudes were not available, attitudes taken from nearby rocks were used. Each sample was individually stepwise demagnetized using

either stepwise alternating field (AF), stepwise thermal, or continuous thermal demagnetization (CTD; *Dunn and Fuller*, [1984]), and measured on a Molspin spinning magnetometer or a 2G cryogenic magnetometer. Stable directions were selected using equal area and *Zijderveld* [1967] vector end point plots. Site averages were then computed using the statistics of *Fisher* [1953].

PALEOMAGNETIC RESULTS FROM THE PHILIPPINE SEA PLATE

The Palau Islands

The Palau Islands are located on the southern end of the Palau-Kyushu ridge and are composed of island-arc subaerial and submarine volcanic rocks and reef limestones ranging in age from upper Eocene to Holocene (Figure 2). *Mason et al.* [1956] established the relative stratigraphic relation between the rock units, which is summarized in Figure 2. Pliocene to Pleistocene reef limestones and a Miocene to Pliocene clay unit overlie a volcanic section which consists of the late Eocene Babeldaup Formation, the early Oligocene Aimeliik Formation, and the Oligocene to Miocene Ngermelengui Formation (containing the Nghemesed, Medorm, and Arakabesan Members). K/Ar age determinations made on the volcanic section [*Meijer et al.*, 1983; *Haston et al.*, 1988] yield an early Miocene age for the Arakabesan Member, a mid-Oligocene age for the Medorm Member, and early Oligocene age for the Aimeliik Formation.

Several paleomagnetic studies have been undertaken in the Palau Islands. *Aoki* [1972] presented preliminary results which show clockwise deflected directions from the early Miocene Arakabesan member of the Ngermelengui Formation. The sites are located in strata which dips up to 30°; however, only in situ directions are cited. Although these data have value as predictors of clockwise rotation, we have not used data from this study because of the lack of structural information. *Adachi et al.* [1987] presented data from 11 sites from Eocene and

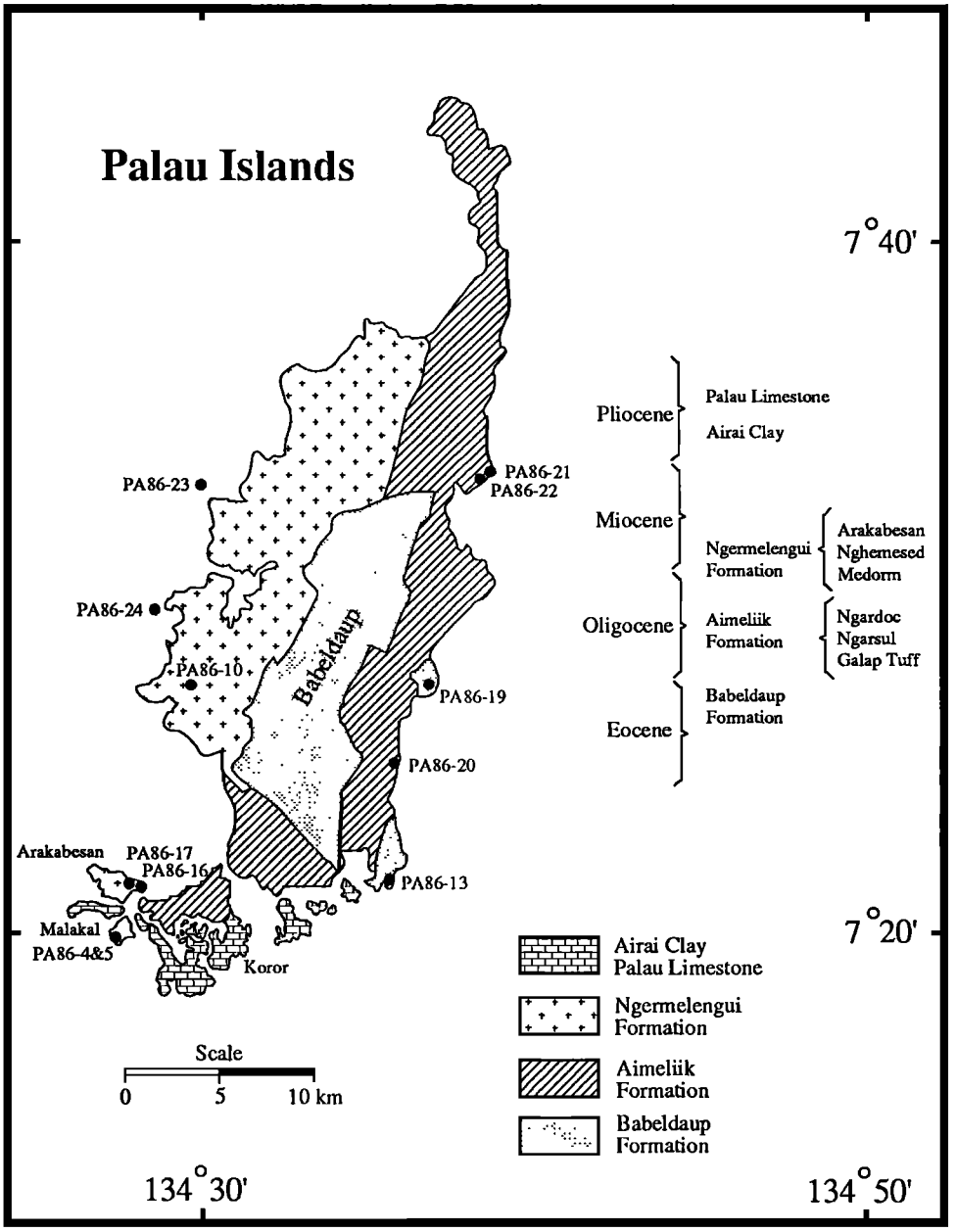


Fig. 2. Generalized geologic map of the Palau Islands with paleomagnetic sampling sites [after *Mason et al.*, 1956].

Oligocene age rocks. No distinction is made between the different age rocks. Eight of their sites show clockwise deflected directions. Although their results agree with the results presented by *Haston et al.* [1988], it is difficult to evaluate the quality of these results because only a cursory discussion has been given. No numerical data are cited. *Haston et al.* [1988] presented data from 12 sites located in the volcanogenic rocks (Figure 2). Figure 3*a* and Table 1 show the results of 5 sites from the lower Miocene Arakabesan Member; the sites have an average direction which is 55° clockwise deflected (I=3.5°, D=54.5°, $\alpha 95$=17.4°, k=20.3). Two sites (PA86-10, PA86-23) from the middle Oligocene Medorm Member show further clockwise deflected directions. Two sites (PA86-13, PA86-19) from the late Eocene Babeldaup Formation have similar directions as the middle Oligocene sites. These two sites are located in heavily mineralized and altered basaltic tuffs and flows and have been interpreted as middle Oligocene remagnetization directions based on rock magnetic evidence [*Haston et al.*, 1988; *Haston*, 1988]. These four sites give an average middle Oligocene direction of

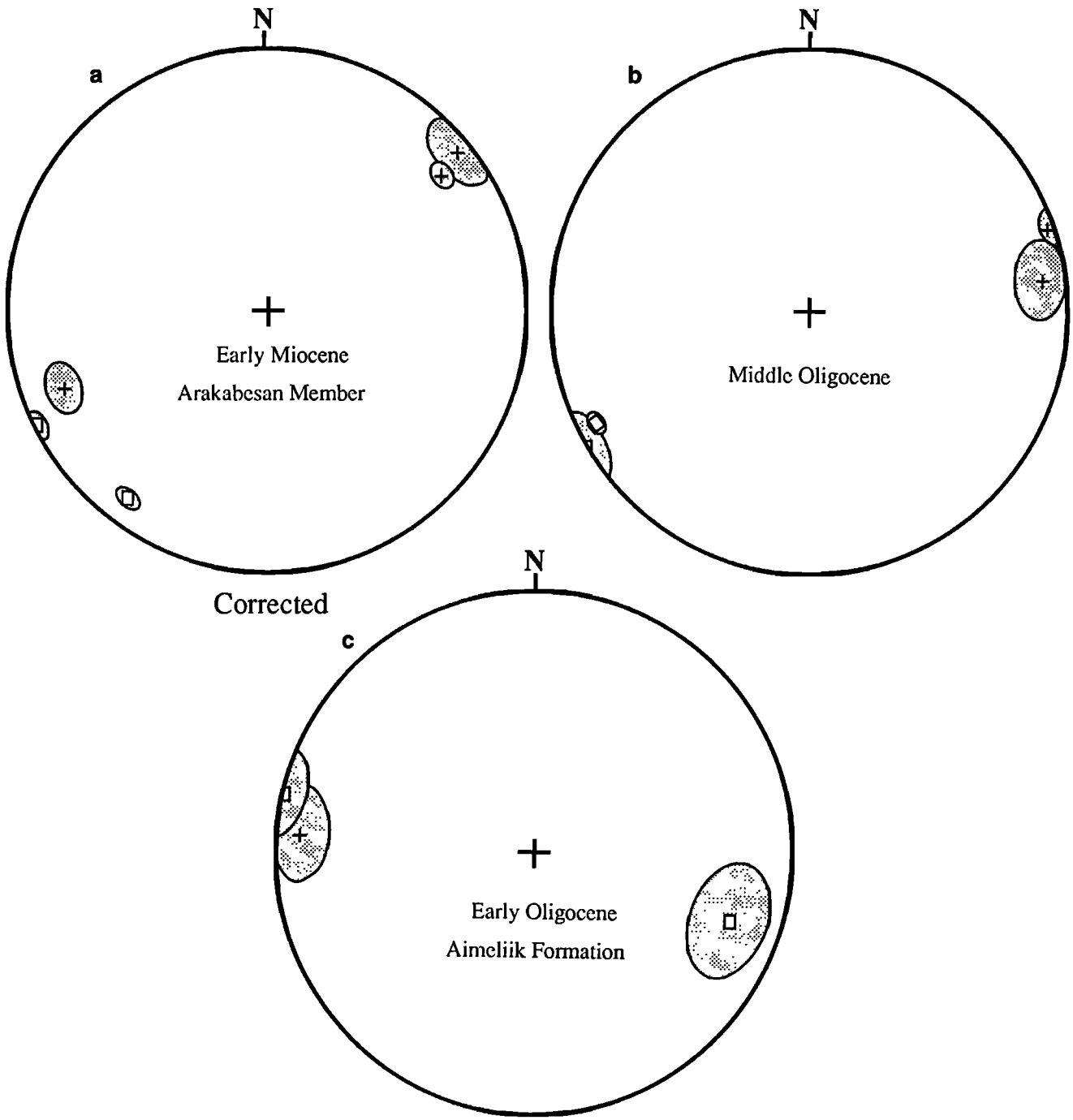


Fig. 3. Equal-area stereographic projection of the site mean directions and associated circles of 95% confidence from the Palau Islands. Pluses indicate a lower hemisphere mean; open squares indicate a upper hemisphere mean. (a) Corrected site mean directions from the early Miocene Arakabesan Member. (b) Interpreted middle Oligocene site mean directions. The two sites from the Medorm Member are in corrected coordinates; the two remagnetized sites from the Babeldaup Formation are in in situ coordinates. (c) Early Oligocene site mean directions from the Aimeliik Formation. Site PA86-21&22 is interpreted as a remagnetization direction, and is in in situ coordinates; the other two sites are in corrected coordinates.

I=5.3°, D=68.4°, α95=12.8°, k=52.4 [Figure 3*b*, Table 2). Three sites from the early Oligocene Aimeliik Formation are strongly (>90°) clockwise deflected (Figure 3*c*, Table 3) and were initially interpreted as local rotation associated with the initial opening of the Parece-Vela basin [Haston et al., 1988]. However, remagnetization and poor magnetic stability was found in many of the samples, thus making the data unreliable.

Guam

Guam lies on the southern end of the Mariana island arc. The geology of Guam was first mapped by *Tracey et al.* [1964] and is summarized in Figure 4. They defined four major stratigraphic units: the Neogene reef limestones, and the volcanogenic Umatac, Lower Facpi, and Alutom Formations. They assigned an Oligocene or Eocene age to the Alutom Formation, and a Miocene age to the Umatac and Lower Facpi Formations. However, subsequent mapping [*Reagan and Meijer*, 1984] and radiometric age dating [*Meijer et al.*, 1983] indicate that the lower Oligocene/upper Eocene Lower Facpi Formation is older than the middle Oligocene Alutom Formation.

Review of past work. *Kobayashi et al.* [1971] collected three reconnaissance hand samples from the upper Eocene Facpi Formation, which at the time was incorrectly thought to be of Miocene age. The three samples show a clockwise deflected direction around 80°. The preliminary nature of the study makes the data unsuitable for use here. *Larson et al.* [1975] collected eight sites from basalt pillows and dikes of the Facpi Formation. AF demagnetization to only 20 mT was used to demagnetize the rocks, which based on our studies of

TABLE 1. Early Miocene Results From the Palau Islands

Site			In Situ		Bedding	Corrected			
Name	Rock Type	*N*	Inc.	Dec.	Correction	Inc.	Dec.	a95	k
PA86-4	basalt dike	7	-21.6	49.8	170/32 W	5.7	50.8	9.1	44.8
PA86-5	basalt sill	7	-18.5	52.3	170/32 W	16.2	52.8	3.0	391.3
PA86-16	basalt flow	7	18.9	246.0	194/41 W	-2.1	243.4	3.1	403.6
PA86-17	basalt flow	10	39.3	257.2	194/41 W	18.1	249.1	6.5	57.0
PA86-24	basalt dike	6	11.2	216.5	098/25 S	-10.8	216.5	3.3	404.7

		In Situ				Corrected			
	N	Inc.	Dec.	a95	k	Inc.	Dec.	a95	k
Mean direction	5	-22.4	55.4	16.9	21.4	3.5	54.5	17.4	20.3

Inc., inclination in degrees; Dec., declination in degrees; a95, circle of 95% confidence about the site mean; In situ, site mean direction before application of tectonic correction; Corrected, site mean direction after tectonic correction.

TABLE 2. Middle Oligocene Results From the Palau Islands

Site			In Situ		Bedding	Corrected			
Name	Rock Type	*N*	Inc.	Dec.	Correction	Inc.	Dec.	a95	k
PA86-10	massive tuff	6	-26.0	250.4	110/30 S	-8.3	242.6	2.4	784.9
PA86-23	andesite flow	7	8.9	67.7	055/32 S	2.1	70.7	4.5	181.6
PA86-13*	altered tuff	7	10.2	81.9	218/22 W	24.2	76.6	9.9	38.4
PA86-19*	basalt flow	6	-0.3	238.8	105/25 S	-18.2	241.4	8.2	67.3

		In Situ			
	N	Inc.	Dec.	a95	k
Mean direction	4	5.3	68.4	12.8	52.4

Abbreviations are as in Table 1.

* Remagnetized late Eocene aged rocks. In situ directions averaged with corrected directions of other two sites.

TABLE 3. Early Oligocene Results From the Palau Islands

Site			In Situ		Bedding	Corrected			
Name	Rock Type	*N*	Inc.	Dec.	Correction	Inc.	Dec.	a95	k
PA86-20	massive tuff	13	-46.1	94.1	228/36 N	-20.9	109.6	15.4	8.1
PA86-22	bedded tuff	5	42.8	257.3	218/45 W	9.6	273.8	11.6	44.5
PA86-21&22*	bedded tuff	6	-0.2	282.4	218/45 W	-34.8	275.0	9.8	48.0

		In Situ			
	N	Inc.	Dec.	a95	k
Mean direction	4	5.3	68.4	12.8	52.4

Abbreviations are as in Table 1.

* Early Oligocene remagnetization direction. In situ direction averaged with other two sites.

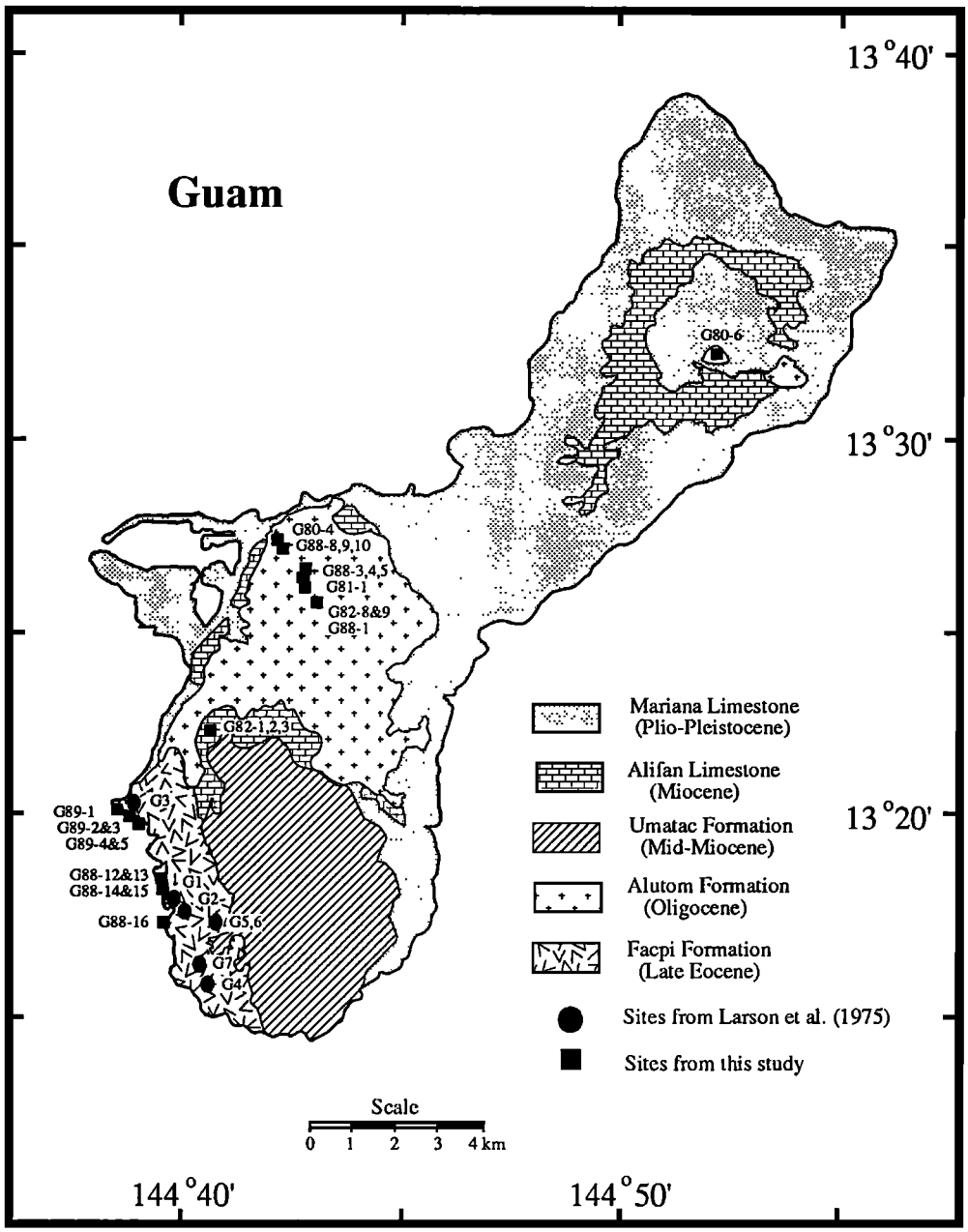


Fig. 4. Generalized geologic map of Guam with paleomagnetic sampling sites [after *Tracey et al.*, 1964].

the formation is generally sufficient to isolate a single component of magnetization. Stepwise thermal demagnetization to 325°C was also used on site G8, although no consistent remanent magnetization direction was obtained. Of the eight sites, seven yield characteristic directions which we considered minimally reliable. No structural corrections are cited, although this may be because of the lack of structural indicators at the sites. The directions are clockwise deflected between 12° and 87° from north, and are summarized in Table 6. *Keating and Helsley* [1985] present data from 19 sites taken from the various formations on Guam. In general the results show clockwise deflected directions. However, only declination data in the form of graphical plots are presented. No structural information, inclinations, in situ directions, demagnetization treatment, or errors are cited, making evaluation of the data impossible. Although these data agree with the overall trend of clockwise rotation, no data from this study have been used for the above reasons.

New data from Guam. New sites were collected from Pliocene to Eocene rocks on Guam; 26 of these sites yielded reliable characteristic remanent magnetic directions. These data will be discussed from youngest to oldest. Eight sites were collected from the Plio-Pleistocene Mariana limestone. No results were obtained because of the rocks' very weak magnetism (between 5×10^{-9} and 6×10^{-7} Am^2/kg) coupled with unstable demagnetization behavior. Three sites from the Miocene Alifan Limestone yield characteristic remanent magnetization directions (Figure 5*a*) which are indistinguishable from the present reversed field direction (Figure 6*a*; Table 4). These rocks have natural remanent magnetization (NRM) intensities around 10^{-5} Am^2/kg, which suggests that the magnetization is not primary. The blocking temperature for these rocks is ≈500°C and suggests that magnetite is the magnetic carrier. The lack of any dip in the beds makes determination of the age of magnetization impossible.

Twelve sites collected from flows and sediments of the

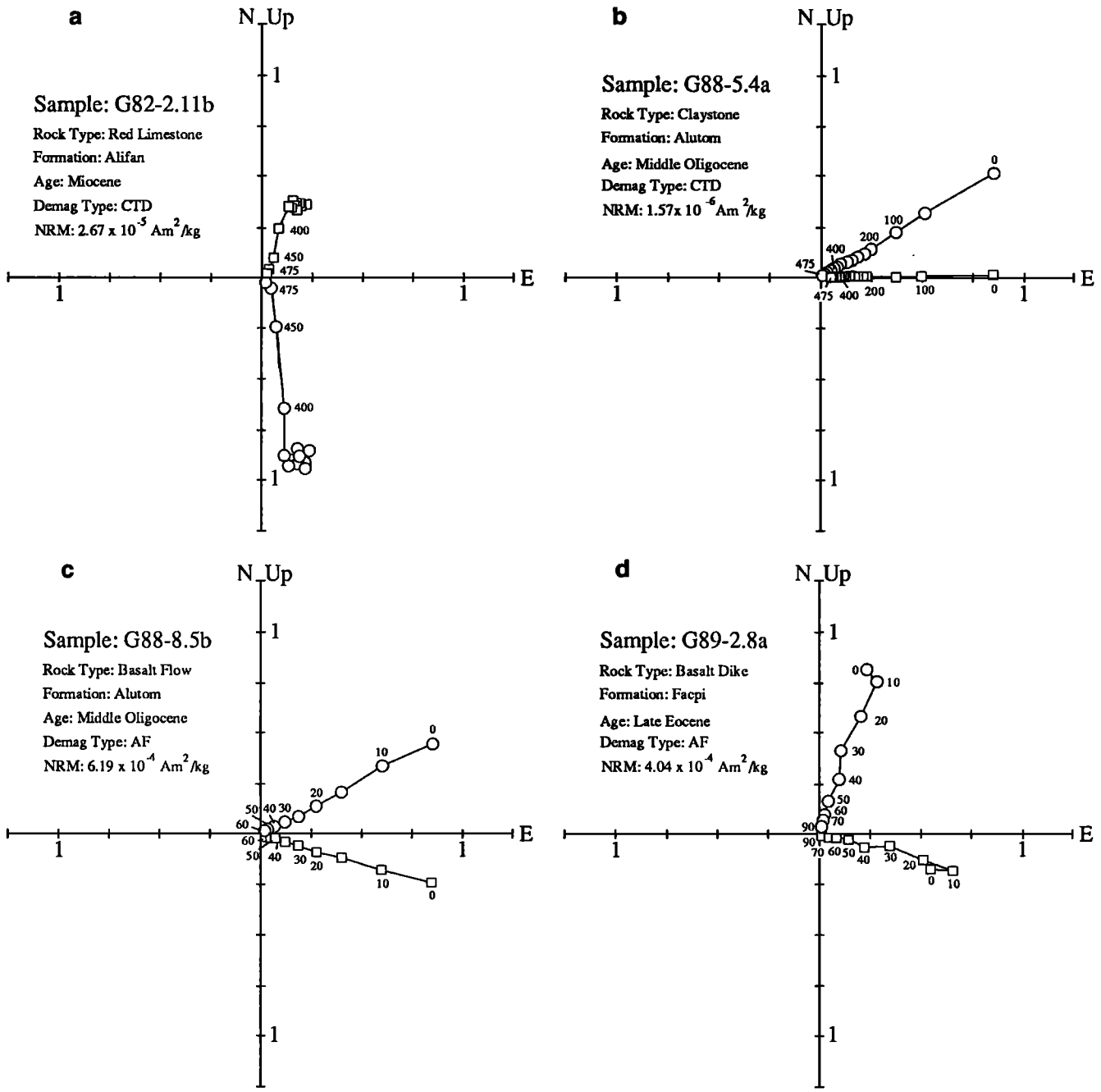


Fig. 5. *Zijderveld* [1967] vector endpoint plots for representative samples from Guam. Squares are the vector projection onto the declination plane; circles are the vector projection onto the inclination-north plane. (a) Continuous thermal demagnetization of sample G82-2.11b from the Miocene Alifan Limestone Formation. (b) Continuous thermal demagnetization of sample G88-5.4a from the middle Oligocene Alutom Formation. (c) AF demagnetization of sample G88-8.5b from the middle Oligocene Alutom Formation. (d) AF demagnetization of sample G89-2.8a from thelate Eocene Facpi Formation.

Oligocene Alutom Formation yield consistent magnetic directions (Figures 5*b* and 5*c*; Table 5). Three of these sites (G80-6, G80-1, G80-8) show directions close to the present field, but slightly counter-clockwise deflected (Figure 6b). The remaining nine sites show strongly clockwise deflected directions (Figure 6*c*). Three sites (G88-3, G88-4, G88-5), collected along a shallowly plunging anticline, pass a fold test at the 95% confidence level [*McFadden and Jones*, 1981]. The dispersion parameter k improves from 12.1 to 20.1 with application of structural corrections, suggesting that the magnetization is at least pre-folding in age. The mean of the corrected directions is I=15.1°, D=66.1° (α95=11.1°, k=20.1).

Ten new sites were collected from the late Eocene Facpi Formation exposed on the southern portion of Guam. Samples were collected from basalt flows, dikes, and pillows. In places, contacts between units appear to be dipping as much as 45°, although the attitudes of the dikes cutting the pillow lavas and flows are subvertical. In general, very few structural indicators are present. Thus structural corrections for these units are not very reliable. In both in situ and corrected coordinates, all of the sites show clockwise deflected directions which range from 18° to 92° (Figure 6*d*; Table 6). The inclinations of the sites are also widely scattered and range between -32° and 52°. The data presented from the Facpi Formation by *Larson et al.*

[1975] show a similar amount of scatter. There is no correlation of geographic location or rock type (i.e., dikes, flows) with direction, indicating that the scatter in the data cannot be solely caused by lack of structural control or different age units. The NRM demagnetization of the samples is generally single component (Figure 5*d*) with high narrow blocking temperature spectra (between 540° and 580°C), suggesting that the magnetization is primary and carried by magnetite. Although all the data show clockwise deflection and are nonrandom at the 95% confidence level, we reject the result from further tectonic analysis because of the large scatter in the site directions. It is not clear why the results show such extreme scatter; it may be due to a combination of secular variation and poor structural control.

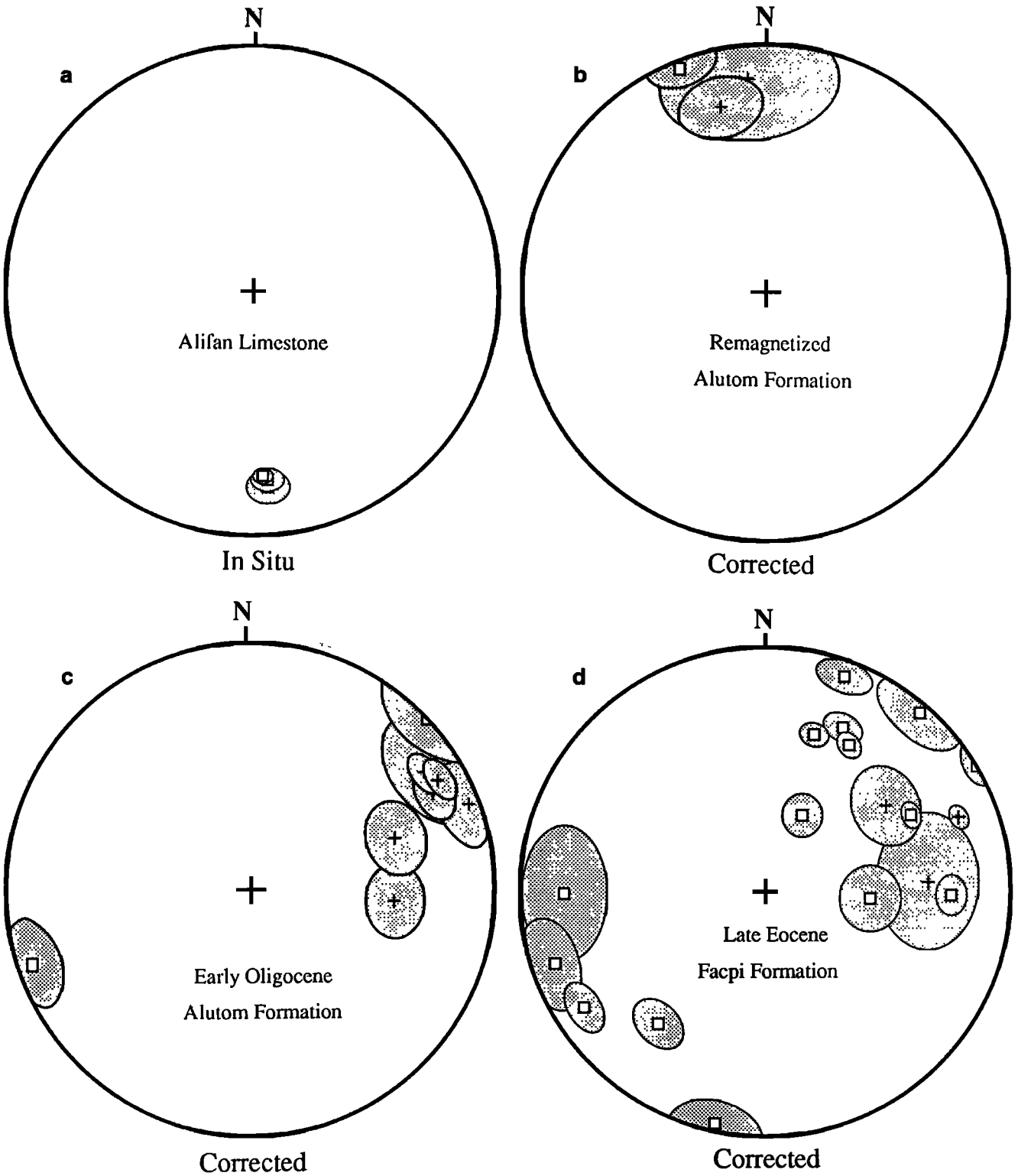


Fig. 6. Equal-area stereographic projection of the site mean directions and associated circles of 95% confidence from Guam. Pluses indicate a lower hemisphere mean; open squares indicate an upper hemisphere mean. (a) In situ site mean directions from the Miocene Alifan Limestone Formation. (b) Remagnetized site mean directions from the Alutom Formation in in situ coordinates.c) Corrected site mean directions from the early Oligocene Alutom Formation. (d) Corrected site mean direction from the late Eocene Facpi Formation.

TABLE 4. Site Results From the Miocene Alifan Formation, Guam

Site Name	Rock Type	N	In Situ Inc.	In Situ Dec.	Bedding Correction	Corrected Inc.	Corrected Dec.	a95	k
G82-1	limestone	6	-25.2	176.4	horizontal	-25.2	176.4	2.8	590.8
G82-2	limestone	6	-20.7	175.5	horizontal	-20.7	175.5	5.9	129.3
G82-3	limestone	5	-23.9	175.1	horizontal	-23.9	175.1	4.1	346.3

	N	In Situ Inc.	In Situ Dec.	a95	k
Mean direction	3	-23.1	175.7	4.2	877.5

Abbreviations are as in Table 1.

TABLE 5. Site Results From the Oligocene Alutom Formation, Guam

Site Name	Rock Type	N	In Situ Inc.	In Situ Dec.	Bedding Correction	Corrected Inc.	Corrected Dec.	a95	k
G80-4	basalt flow	5	14.1	51.7	072/37 N	-0.3	47.3	13.1	34.7
G80-6*	basalt flow	4	14.1	354.5	183/57 W	---	---	23.7	16.1
G81-1*	mudstone	5	-1.8	338.8	083/4 S	---	---	8.6	79.4
G82-8*	mudstone	4	23.4	345.7	090/30 S	---	---	11.9	60.6
G82-9	mudstone	5	-5.9	257.2	072/37 N	-7.1	252.5	11.4	45.9
G88-1	mudstone	4	11.8	59.3	223/7 N	13.7	57.7	15.8	34.6
G88-3	mudstone	6	45.4	120.2	297/25 N	40.5	96.8	11.2	36.5
G88-4	mudstone	6	52.7	71.1	345/16 E	36.9	72.6	11.5	34.7
G88-5	mudstone	6	-0.8	68.2	083/28 S	6.5	70.1	11.1	37.5
G88-8	basalt flow	7	7.1	65.4	232/25 N	11.8	61.5	6.9	75.5
G88-9	basalt flow	7	11.1	64.5	232/25 N	14.9	58.7	6.6	83.2
G88-10	basalt flow	8	9.7	68.9	232/25 N	16.2	63.5	8.4	44.3

	N	In Situ Inc.	In Situ Dec.	In Situ a95	In Situ k	Corrected Inc.	Corrected Dec.	Corrected a95	Corrected k
Mean direction	9	16.5	70.1	14.5	12.1	15.1	66.1	11.1	20.1

Abbreviations are as in Table 1.
* Remagnetized Plio-Pleistocene direction.

TABLE 6. Site Results From the Eocene Facpi Formation, Guam

Site Name	Rock Type	N	In Situ Inc.	In Situ Dec.	Bedding Correction	Corrected Inc.	Corrected Dec.	a95	k
G88-12	pillow basalt	6	32.8	55.1	353/37 E	-0.7	59.2	5.2	162.6
G88-13	basalt dike	6	12.2	91.3	353/37 E	-25.4	91.5	5.5	149.6
G88-14	basalt flow	7	-33.7	55.6	353/37 E	-62.1	26.7	6.8	78.6
G88-15	pillow basalt	5	51.7	61.1	353/37 E	16.5	69.8	3.2	581.1
G88-16	pillow basalt	6	8.6	20.1	353/37 E	-7.2	20.5	7.2	88.5
G89-1	basalt dike	7	11.3	27.9	280/45 N	-32.2	30.3	4.0	227.6
G89-2	basalt dike	7	10.6	17.1	280/45 N	-34.2	17.9	4.0	229.9
G89-3	basalt flow	7	-30.6	54.5	280/45 N	-54.3	93.7	10.0	37.1
G89-4	basalt dike	7	16.4	24.6	280/45 N	-27.7	25.6	5.7	112.6
G89-5	pillow basalt	7	-0.8	52.6	280/45 N	-33.7	63.1	3.3	328.1
G80-1	basalt	5	-8.4	270.6	230/12 N	-16.1	268.6	17.1	20.9
G1*	NA	9	33.0	87.0	NA	----	----	20.0	8.0
G2*	NA	8	40.0	56.0	NA	----	----	12.0	22.0
G3*	NA	19	-4.0	41.0	NA	----	----	12.0	9.0
G4*	NA	16	-31.0	218.0	NA	----	----	8.0	20.0
G5*	NA	6	-2.0	192.0	NA	----	----	11.0	35.0
G6*	NA	6	-13.0	237.0	NA	----	----	7.0	91.0
G7*	NA	7	-8.0	251.0	NA	----	----	12.0	25.0

	N	In Situ Inc.	In Situ Dec.	In Situ a95	In Situ k	Corrected Inc.	Corrected Dec.	Corrected a95	Corrected k
Mean direction	18	13.5	52.4	14.8	6.8	-7.8	55.3	18.9	4.5

NA means data not cited or not available; other abbreviations are as in Table 1.
* from Larson et al. [1975].

Saipan

Saipan lies 200 km northeast of Guam, along the Mariana island arc. The geology of Saipan was first mapped by *Cloud et al.* [1956]. They describe the island as a volcanic core enveloped by reef limestones. The rocks on Saipan have been divided into five major formations: the Sankakuyama, the Hagman, the Fina Sisu, the Tagpachau Limestone, and the Mariana Limestone (Figure 7).

The Sankakuyama Formation is dominated by dacitic volcanic flows and tuffs, which are poorly exposed in small fault bounded blocks in the central and northern portions of the island. The formation yielded late Eocene whole-rock and feldspar-separate K/Ar dates [*Meijer et al.*, 1983]. Unconformably overlying the Sankakuyama Formation is the Hagman Formation. The Hagman Formation is dominated by water-laid volcanogenic sediments and andesitic pyroclastic flows, but sparse andesitic flows and flow breccias are present. A K/Ar date from the formation suggests an early Oligocene age (35.7 ± 0.5 Ma.; *Meijer et al.* [1983]), which is in conflict with the upper Eocene fossil age [*Cloud et al.*, 1956]. Overlying the older volcanic rocks are rocks of the Tagpachau Limestone and Fina Sisu Formations. The Tagpachau Limestone Formation, exposed throughout the island, is dominantly reef limestones Oligocene and Miocene in age [*Cloud et al.*, 1956]. The Fina Sisu Formation comprises basaltic flows and pyroclastics, which are poorly exposed in the southern portion of the island. A middle Miocene K/Ar date of 12.0 ± 0.3 Ma has been obtained from this formation [*Meijer et al.*, 1983]. The youngest rocks on Saipan are the well-exposed Plio-Pleistocene reef limestones of the Mariana Limestone Formation.

Review of past work. *Fuller et al.* [1980] were the first to report paleomagnetic data from Saipan. Their results suggested that Saipan has undergone significant clockwise rotation since the late Eocene. However, the data are sparse and some sites were misreported, as will be discussed later. *Keating and Helsley* [1985] also reported paleomagnetic data from Saipan that suggested that Saipan has undergone progressive clockwise rotation since the Miocene. However, it is difficult to evaluate the quality of their data, because only declination data in the form of graphical plots are presented. No structural information, inclinations, in situ directions, demagnetization treatment, or errors are described.

New Paleomagnetic results. Samples were collected from a wide variety of rocks ranging from Plio-Pleistocene to Eocene

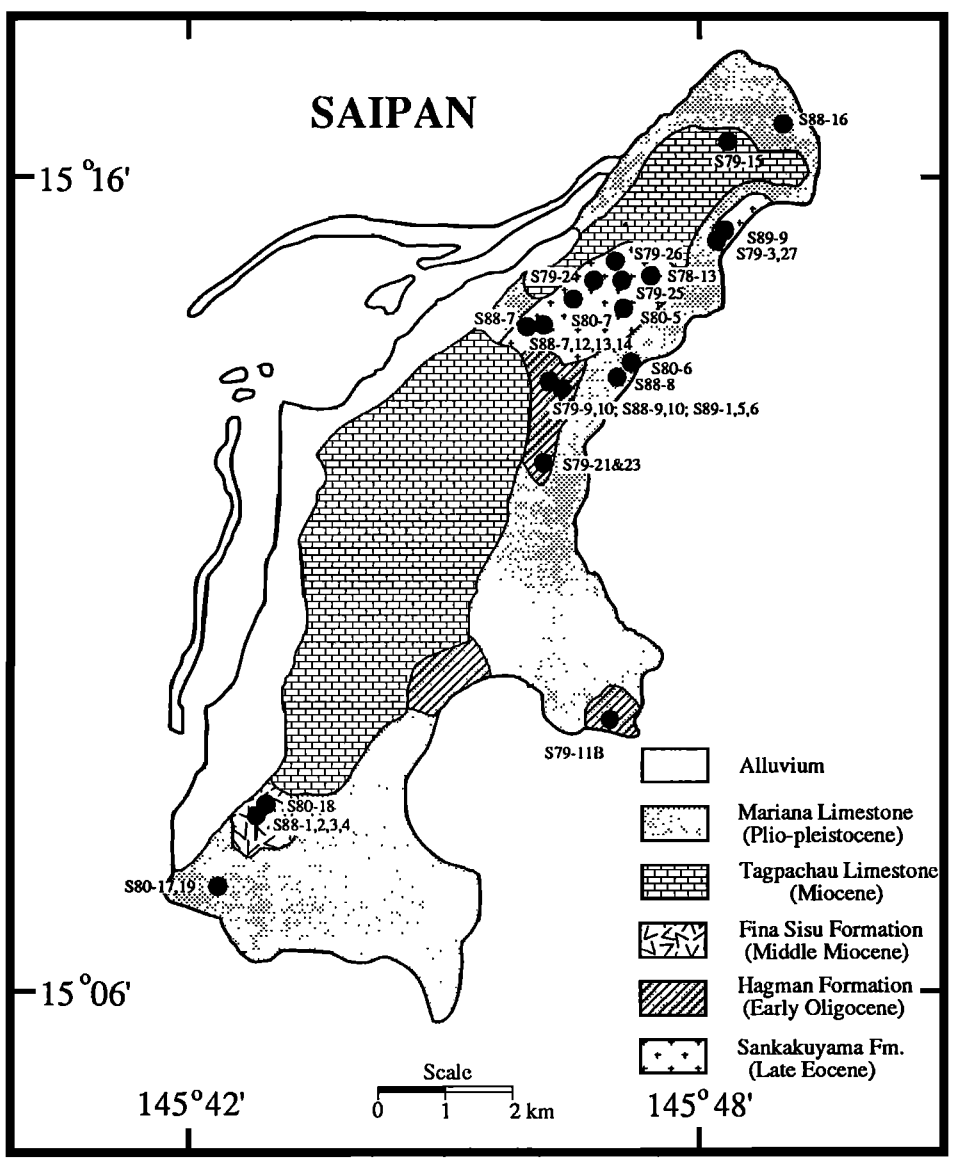


Fig. 7. Generalized geologic map of Saipan with paleomagnetic sampling sites [after *Cloud et al.*, 1956].

in age, with many of the sites yielding characteristic magnetic directions. These data will be discussed from youngest to oldest for ease of understanding. Twelve sites were collected from the Plio-Pleistocene Mariana Limestone Formation, although the low magnetic intensity and unstable demagnetization behavior made most of the sites unusable. Four sites yield consistent remanent magnetic directions (Table 7; Figure 9*a*), and have a mean direction of I=8.6°, D=350.1° (α95=22.5°, k=17.5), which is deflected slightly counter-clockwise from the present field direction. However, the large α95 makes the result indistinguishable from the present dipole field.

Four new sites collected from basaltic flows of the middle Miocene Fina Sisu Formation yield characteristic remanent magnetic directions. The extreme weathering of the rocks make exposure of the Fina Sisu Formation very poor. Thus structural attitudes in the area are difficult to obtain; attitudes where taken from contacts between visible flow units. A characteristic direction was easily determined in almost all of the samples (Figure 8*a*). These four sites yield similar clockwise deflected directions as the one site reported by *Fuller et al.*, 1980 (Figure 9*b*; Table 8).

Although many sites were collected from the Miocene Tagpachau Limestone Formation, only one site yields a consistent remanent direction. The other sites were either magnetically unstable (Figure 8*b*) or too weak to measure (between 5 x 10^{-9} and 5 x 10^{-8} Am^2/kg). The stable results are clockwise deflected a similar amount to the Fina Sisu results (Figure 9*b*; Table 8). The mean Middle Miocene direction for Saipan is I=-30.7°, D=208.1° (α95=8.4°, k=64.6).

Past results from the early Oligocene Hagman Formation [*Fuller et al.*, 1980; *McCabe and Uyeda*, 1983; *Keating and Helsley*, 1985] suggest around 40°-50° of clockwise deflection; however, the quality of the results is very poor. In an effort to better constrain the early Oligocene direction for Saipan, seven new sites were collected from this formation. More sampling was done (sites S88-9, S88-10, S89-1, S89-5, S89-6) in an area where site results had been previously reported. The poor exposure in this area left some question as to whether the particular outcrops were basaltic flows or flow breccias. The individual samples have easily defined single components of magnetization. However, the sample directions, as well as the site mean directions, are extremely scattered (Figure 9*c*, Table 9). This indicates that the outcrop is probably a flow breccia, which was brecciated after the magnetization of the rocks was acquired, and thus sites S79-9, S79-10, S88-9, S88-10, S89-1, S89-5, and S89-6 must be rejected from any tectonic analysis. Of the remaining sites, site S79-11B is indistinguishable from the present dipole field direction, and sites S79-21 and S79-23 are widely scattered. Therefore no reliable results exist from the Hagman Formation.

Fuller et al. [1980] reported four sites from dacite flows of the Late Eocene Sankakuyama Formation which have clockwise deflected declinations between 9° and 46°. Unfortunately, the negative sign of the inclination from one site (S79-26) was lost during printing, thus the reported direction I=46.6° D=42.8° should have read I=-46.6° D=42.8°. To make matters worse, *McCabe and Uyeda* [1983] misreported the inclinations as positive from three additional sites (S79-25, S80-5, S80-7); these sites all have directions with negative inclinations. In addition to these previously reported results, characteristic remanent magnetic directions were obtained from six new sites. These rocks behaved well under both AF and CTD treatment; Figures 8*c* and 8*d* are typical plots of their demagnetization behavior. All of the data from the Sankakuyama Formation are shown in Table 10 and Figure 9*d*. Two of the reported sites (S79-3 and S79-27) have in situ directions which are indistinguishable from the present dipole field and are rejected from tectonic analysis. One site (S88-7) yields an anomalous result from the other sites and is not used in computing the mean direction. We have no justification for rejecting this site except that it is very different from the 10 other site results. However, inclusion of this site in the calculation of the mean direction does not significantly change the mean direction.

The mean in situ direction for the Sankakuyama Formation is I=-10.3°, D=40.9° (α95=18.5°, k=9.8), which is significantly clockwise deflected from the present dipole field position. Many of the sites come from flow-banded dacites located in poorly exposed fault blocks several hundred meters in size. Thus determination of bedding attitude in many of the flows was difficult or impossible. Table 10 indicates which sites have bedding corrections that are somewhat suspect. Field observations indicate that most of the sampled fault blocks have fairly shallow dips. The inclination values are less reliable than the declination, because they are more sensitive to any error in the application of a structural correction. The mean corrected direction (I=-12.0°, D=43.0°, α95=12.5°, k=16.0) is not significantly different from the in situ direction; however, the dispersion in the Sankakuyama site results decreases from k=9.8 to k=16.0 after application of structural corrections, suggesting that the magnetization may be prefolding and that our estimates of bedding are broadly acceptable.

The Bonin Islands

The Bonin Islands, located along the Bonin Ridge, consist of three main groups of islands that are approximately 50 km apart. These are, from south to north, Hahajima, Chichijima, and Mukojima (Figure 10). The Bonin Islands are predominantly made up of volcanic rocks thought to have formed along an active island arc, prior to the opening of the

TABLE 7. Site Results From the Plio-Pleistocene Mariana Limestone, Saipan

Site			In Situ		Bedding	Corrected			
Name	Rock Type	N	Inc.	Dec.	Correction	Inc.	Dec.	a95	k
S80-6*	limestone	5	20.8	347.2	horizontal	20.8	347.2	5.5	196.3
S80-17&19	limestone	8	14.1	354.2	horizontal	14.1	354.2	6.5	73.8
S88-8	limestone	5	11.4	4.7	horizontal	11.4	4.7	4.1	349.1
S88-16	limestone	5	12.9	154.0	horizontal	12.9	154.0	5.2	219.3

		In Situ			
	N	Inc.	Dec.	a95	k
Mean direction	4	8.6	350.1	22.5	17.5

Abbreviations are as in Table 1.

* Reevaluated from Fuller et al. [1980].

Parece-Vela and Shikoku basins. K-Ar ages from around the islands suggest that the rocks are late Eocene in age [*Tsunakawa*, 1983]. The Bonin Islands are the type locality for the high MgO boninites.

Several paleomagnetic studies have been reported for the Bonin Islands [*Kodama*, 1981; *Keating et al.*, 1983; *Kodama et al.*, 1983; *Keating and Helsley*, 1985]. No new data are presented in this study, but we will summarize all of the published results from the islands.

Chichijima. Chichijima is composed of mainly boninitic and andesitic pillow lavas, hyaloclastites, dikes, and subordinate sedimentary rocks. The dips of the volcanic units are generally gentle (less than 20°) except in a few areas [*Keating et al.*, 1983]. Dikes are especially common on the eastern portion of the islands and change orientation from a WNW direction in the north to a NNW direction in the south. Lower Eocene radiolaria and nannoplankton fossils are present in the northern portion of the island. A widerange of K-Ar ages have been obtained, with the oldest being late Eocene [*Tsunakawa*, 1983]. *Tsunakawa* [1983] attributes the scatter in the K-Ar ages to a thermal disturbance around 5-10 Ma. The rocks are probably all upper Eocene in age.

A total of 27 sites yield characteristic remanent magnetic directions [*Kodama*, 1981; *Kodama et al.*, 1983; *Keating et al.*, 1983]; these data are summarized in Table 11 and Figure 11*a*. The majority of the sites show strongly clockwise deflected

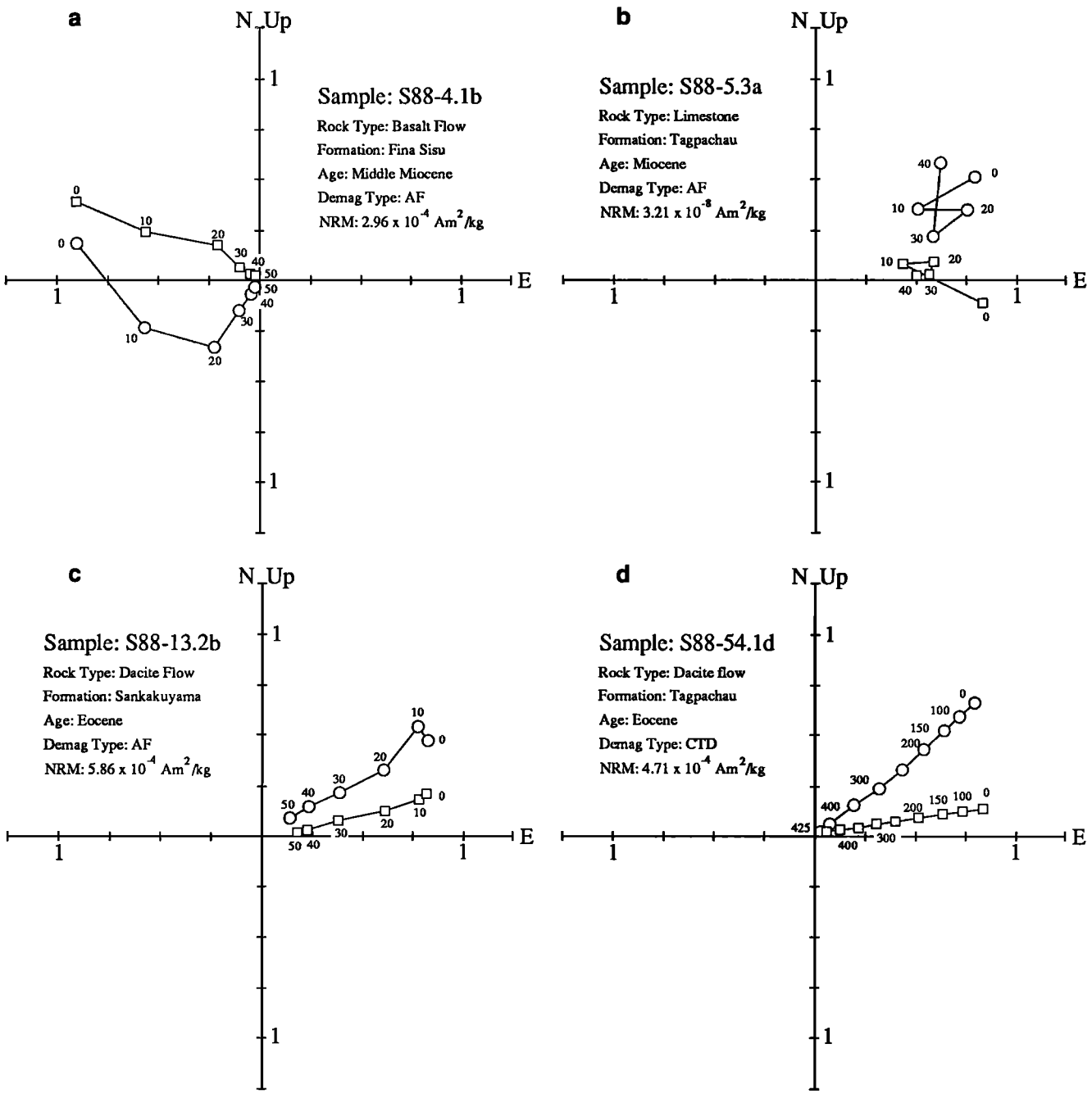


Fig. 8. *Zijderveld* [1967] vector endpoint plots for representative samples from Saipan. Squares are the vector projection onto the declination plane; circles are the vector projection onto the inclination-north plane. (a) AF demagnetization of sample from the Miocene Fina Sisu Formation. (b) AF demagnetization of sample from the Miocene Tagpachau Limestone Formation. (c) AF demagnetization of sample from the late Eocene Sankakuyama Formation. (d) Continuous thermal demagnetization of sample from the late Eocene Sankakuyama Formation.

directions, and the mean in situ direction of the 27 sites is I=9.5°, D=105.3° (α95=12.6°, k=6.1). Of the 27 sites, only 9 have been corrected for structural tilting and 2 are reported as horizontal. A corrected direction of I=9.9°, D=105.6° (α95=11.6°, k=7.0) was calculated by averaging the known corrected directions with the in situ directions from the sites where no tectonic information has been cited. Note that there is very little difference between the mean in situ and corrected directions (Table 11). This, coupled with the fact that the strata are only gently dipping, suggests that the magnetization direction is of late Eocene origin.

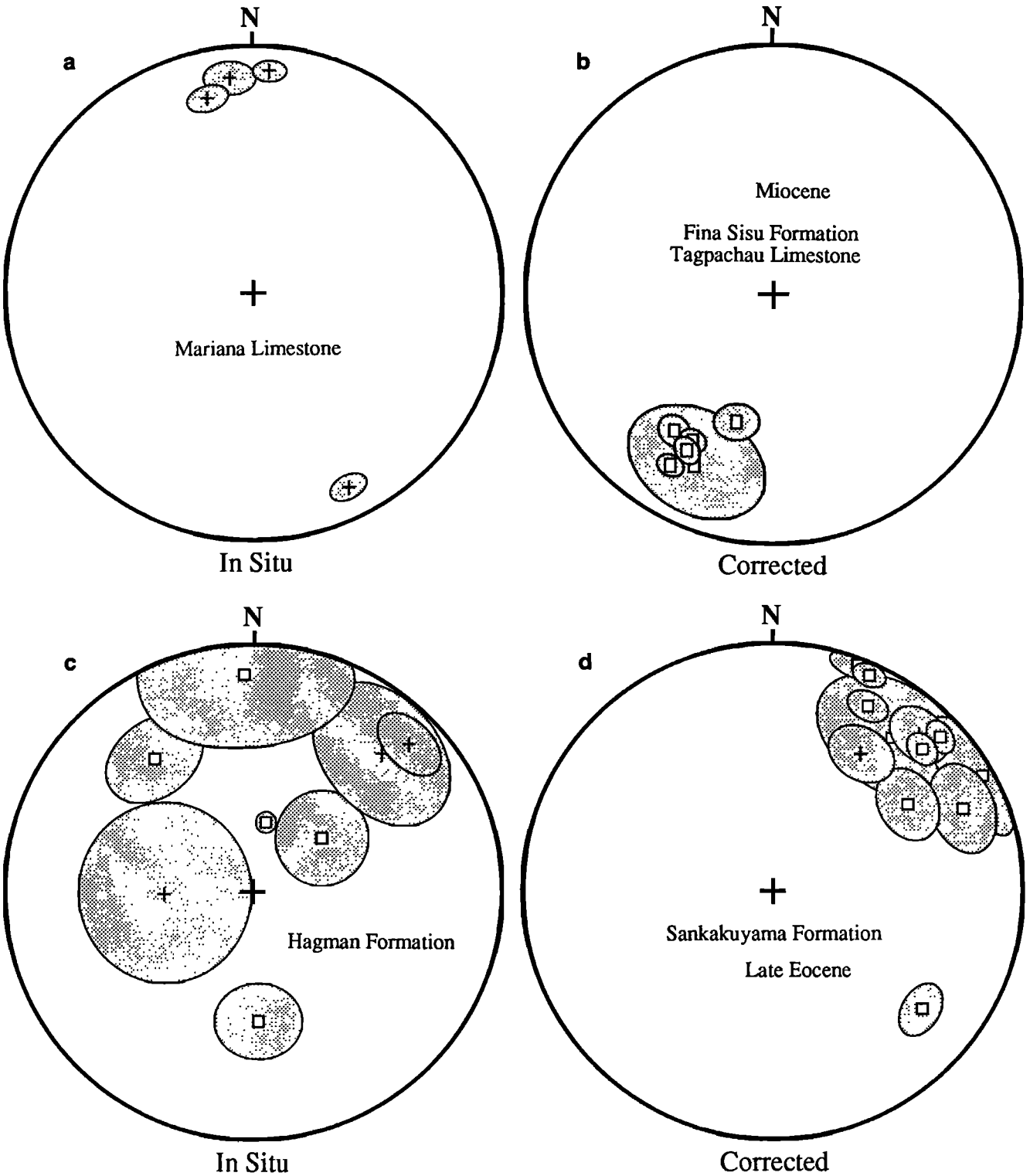


Fig. 9. Equal-area stereographic projection of the site mean directions and associated circles of 95% confidence from Saipan. Pluses indicate a lower hemisphere mean; open squares indicate an upper hemisphere mean. (a) In situ site mean directions from the Plio-Pleistocene Mariana Limestone. (b) Corrected site mean directions from the Miocene Fina Sisu and Tagpachau Limestone Formations. (c) In situ site mean directions from the early Oligocene Hagman Formation. (d) Corrected site mean directions from the late Eocene Sankakuyama Formation.

TABLE 8. Site Results From Miocene Rocks of Saipan

Site Name	Rock Type	N	In Situ Inc.	In Situ Dec.	Bedding Correction	Corrected Inc.	Corrected Dec.	a95	k
				Fina Sisu Formation					
S88-1	basalt flow	7	-47.3	192.4	343/25 E	-31.8	208.9	4.1	228.1
S88-2	basalt flow	7	-45.2	193.2	343/25 E	-29.7	208.6	5.1	145.7
S88-3	basalt flow	7	-50.5	200.1	343/25 E	-32.3	216.4	5.1	140.2
S88-4	basalt flow	6	-37.2	201.1	343/25 E	-19.9	211.4	3.3	406.6
S80-18*	basalt flow	6	---	---	NA	-44.9	196.1	6.4	111.4
				Tagpachau Limestone					
S79-15	limestone	7	-24.9	205.3	horizontal	-24.9	205.3	20.6	9.5

	N	In Situ Inc.	In Situ Dec.	In Situ a95	In Situ k	Corrected Inc.	Corrected Dec.	Corrected a95	Corrected k
Mean direction	6	-41.8	198.4	8.4	65.1	-30.7	208.1	8.4	64.6

Abbreviations are as in Table 1. NA means data not cited or not available.
* From Fuller et al. [1980].

TABLE 9. Site Results From the Early Oligocene Hagman Formation, Saipan

Site Name	Rock Type	N	In Situ Inc.	In Situ Dec.	Bedding Correction	Corrected Inc.	Corrected Dec.	a95	k
S79-9&10	basalt flow	6	21.4	42.6	NA	---	---	22.8	9.6
S79-11B*†	sandstone	5	20.2	9.7	020/20 E	---	---	10.3	56.1
S79-21	basalt flow	4	13.9	46.6	NA	---	---	10.6	76.8
S79-23	basalt flow?	8	-60.4	52.6	350/32 W	-30.3	64.7	15.2	14.3
S88-9	basalt flow?	8	-66.7	11.2	137/38 S	-45.1	47.7	2.6	464.2
S88-10	basalt flow?	6	-46.3	176.9	137/38 S	-41.2	145.1	13.5	25.5
S89-1	basalt flow?	5	-13.3	357.4	137/38 S	-9.1	3.8	28.2	8.3
S89-5	basalt flow?	6	61.2	269.1	137/38 S	28.8	266.8	29.1	6.2
S89-6	basalt flow?	6	-33.2	323.7	137/38 S	-42.9	348.6	14.8	21.5

Abbreviations are as in Table 1. NA means data not cited or not available.
* Remagnetized in the present field.
† Reinterpreted data from Fuller et al., 1980

TABLE 10. Site Results From the Eocene Sankakuyama Formation, Saipan

Site Name	Rock Type	N	In Situ Inc.	In Situ Dec.	Bedding Correction	Corrected Inc.	Corrected Dec.	a95	k
S78-13†	dacite flow	7	31.0	31.0	260/45 N°	-5.3	24.4	4.8	157.3
S79-3*†	dacite flow	6	40.4	6.5	-----	---	---	9.4	67.2
S79-24	dacite flow	4	34.1	46.2	320/52 N°	-17.8	46.7	5.3	306.9
S79-25	dacite flow	4	-22.0	37.7	NA			21.1	19.8
S79-26†	dacite flow	6	-46.6	42.8	328/38 W°	-9.5	47.5	4.8	193.7
S79-27*†	dacite flow	8	37.2	345.6	-----	---	---	12.8	19.5
S80-5	dacite flow	6	-6.0	51.5	115/33 N°	-34.9	57.7	11.1	37.2
S80-7	dacite flow	6	-9.5	24.0	235/16 N°	-17.1	27.4	6.1	119.9
S88-12	red ash	4	-9.5	18.4	180/25 W	-2.6	20.5	6.8	181.1
S88-13	dacite flow	7	-39.7	60.3	180/25 W	-17.2	66.6	12.7	23.5
S88-14	dacite flow	7	-25.7	57.5	180/25 W	-3.9	61.1	15.2	16.6
S88-7	dacite flow	7	-41.7	140.3	180/25 W	-23.9	128.4	7.6	64.1
S89-4	dacite flow	5	-3.1	44.8	180/25 W	14.4	43.1	10.2	56.9
S89-9	dacite dike	4	36.2	24.2	020/10 E°	34.7	32.9	10.3	80.5

	N	In Situ Inc.	In Situ Dec.	In Situ a95	In Situ k	Corrected Inc.	Corrected Dec.	Corrected a95	Corrected k
Mean direction	10	-10.3	40.9	18.5	9.8	-12.0	43.0	12.5	16.0

Abbreviations are as in Table 1. NA means data not cited or not available.
* Remagnetized in the present field.
† Reinterpreted data from Fuller et al. [1980].
° Poorly determined bedding attitude.

Anijima, Nishijima, and Otootojima. Anijima, Nishijima, and Otootojima have geology similar to Chichijima, with the predominant rock types being andesitic and boninitic pillow lavas, dikes and hyaloclastites. The dike swarm observed on the eastern margin of Chichijima continues along the southern and eastern portion of Anijima. One K-Ar age determination from Otootojima yielded a 22 Ma age [*Tsunakawa*, 1983].

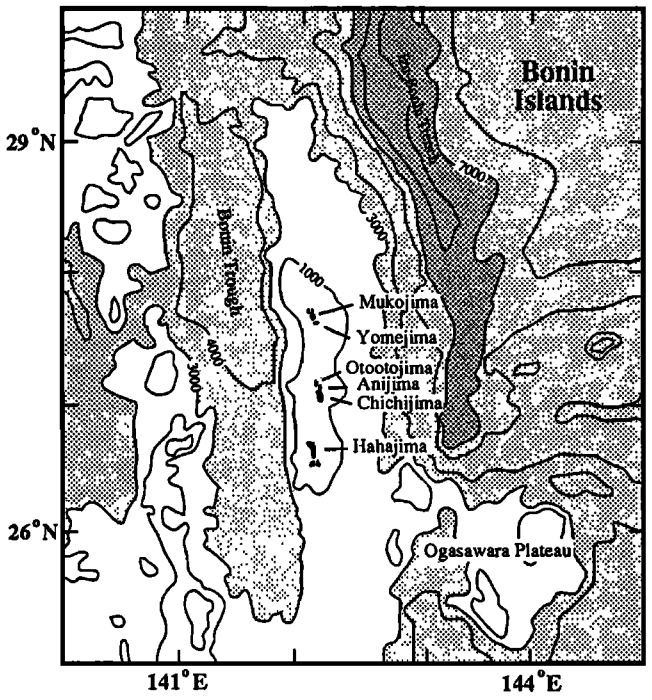


Fig. 10. Mapping showing the location of the Bonin Islands.

However, the similarity of the geology with the other islands suggest that the rocks from Anijima and Otootojima are also upper Eocene in age.

A total of 20 sites were collected from Anijima, 2 from Nishijima, and 6 from Otootojima [*Keating et al.*, 1983]. The results from these sites are summarized in Tables 12 and 13. The sites from Nishijima are averaged together with those from Anijima because of their proximity. Although there is a large amount of scatter, the results predominantly cluster around 090° (Figure 11*b*). A mean in situ direction of I=-0.8°, D=91.7° (α95=16.9°, k=4.9) is obtained from the 22 sites on Anijima and Nishijima. The six sites from Otootojima are widely scattered (α95=46.5°, k=43.0) and cannot be considered reliable.

Hahajima. Hahajima comprises andesitic lavas and volcanic breccias together with a number of dikes. Most of the strata are gently dipping, usually less than 20°, reflecting broadscale warping [*Kodama et al.*, 1983]. Marine sediments contain plankton and foraminifera fossils of late Middle Eocene age [*Saito*, 1962; *Ujiie and Matsumaru*, 1977]. K-Ar dates from the island have a wide scatter and range from 40 Ma. to 9 Ma. [*Kaneoka et al.*, 1970; *Tsunakawa*, 1983]. The similarity in the geology with the other islands suggests that these rocks are also upper Eocene in age.

Only four sites, which yield stable magnetic directions, have been reported from Hahajima [*Kodama*, 1981]. The directions from these sites are scattered, but all show clockwise deflections (Table 13) and yield a mean direction of I=4.7°, D=211.1° (α95=26.4°, k=13.0). This direction is less clockwise deflected than the directions from Anijima and Chichijima; however, the errors nearly overlap at the 95% level of confidence. Further work is needed to determine whether Hahajima has experienced less clockwise rotation than the other islands. We have not included these data because of the large α95 and low number of sites.

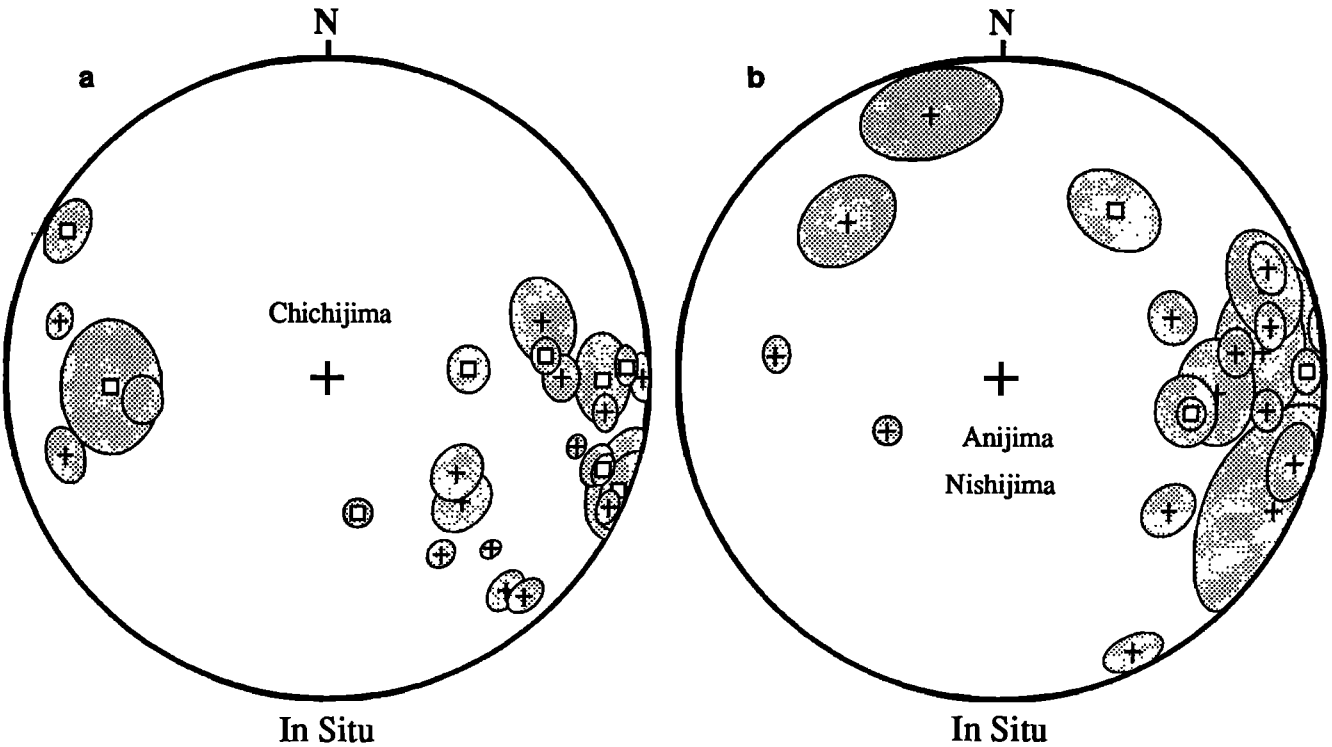


Fig. 11. Equal-area stereographic projection of the site mean directions and associated circles of 95% confidence from the Bonin Islands. Pluses indicate a lower hemisphere mean; open squares indicate an upper hemisphere mean. (a) In situ site mean directions from late Eocene rocks of Chichijima. (b) In situ site mean directions from late Eocene rocks of Anijima and Nishijima.

TABLE 11. Site Results From the Eocene Rocks of Chichijima, Bonin Islands

Site			In Situ		Bedding	Corrected			
Name	Rock Type	*N*	Inc.	Dec.	Correction	Inc.	Dec.	a95	k
LP	andesite	13	-53.7	87.9	NA	---	---	5.1	71.0
ZH	dacite dike	5	47.9	125.8	NA	---	---	6.3	149.0
KM-1	dacite flow	7	-7.2	88.2	horizontal	---	---	3.6	310.0
KM-2	pillow lava	7	-15.2	90.6	NA	---	---	9.2	43.0
MK	pillow lava	6	31.0	75.4	NA	---	---	9.2	53.0
TU	pillow lava	7	27.4	90.4	NA	---	---	5.1	138.0
SD	dacite dike	5	-41.5	261.8	NA	---	---	5.1	243.0
SK	pillow lava	9	-32.5	266.5	horizontal	---	---	14.8	13.0
Site 1	pillow lava	14	13.0	139.8	NA	---	---	4.4	79.8
Site 2	pillow lava	8	-43.9	127.8	NA	-10.3	108.3	2.7	415.7
Site 3	pillow lava	7	12.8	108.2	NA	---	---	4.1	216.0
Site 4-1	andesite dike	9	7.4	137.6	NA	---	---	4.1	157.3
Site 4-2*	andesite dike	7	-53.8	167.2	NA	---	---	3.2	357.1
Site 5	pillow lava	7	2.4	114.6	NA	---	---	3.7	267.6
Site 6	andesite dike	4	16.0	252.7	NA	---	---	5.4	295.2
Site 7	pillow lava	9	13.7	97.5	NA	---	---	3.1	270.5
Site 8	dacite lava	14	30.3	125.3	NA	42.0	132.2	7.5	29.5
Site 9	andesite lava	12	25.5	140.4	NA	34.4	147.4	3.1	199.2
Site 10	dacite dike	4	15.3	132.5	NA	26.0	136.5	1.8	2725.0
Site 11	boninite dike	6	39.9	284.3	NA	15.5	281.3	3.2	438.0
Site 26	andesite dike	15	2.9	110.9	NA	-0.7	111.1	7.9	24.3
Site 27-1	andesite dike	6	-2.7	110.0	NA	---	---	10.0	45.6
Site 27-2	andesite dike	4	-31.9	84.8	NA	---	---	3.5	685.3
Site 27-3	pillow lava	13	60.3	149.5	NA	20.4	105.6	2.1	424.6
Site 28	dacite lava	14	-7.5	299.1	NA	---	---	6.5	38.7
Site 30-1	pillow lava	8	24.5	79.5	NA	23.2	103.3	7.8	52.0
Site 30-2	pillow lava	11	1.3	88.6	NA	1.9	90.1	4.1	133.0

		In Situ				Corrected			
	N	Inc.	Dec.	a95	k	Inc.	Dec.	a95	k
Mean direction	27	9.5	105.3	12.6	6.1	9.9	105.6	11.6	7.0

Data from Kodama [1981], Kodama et al. [1983], Keating et al. [1983]. Abbreviations are as in Table 1. NA means data not cited or not available.

* Interpreted as remagnetized in reverse present field.

TABLE 12. Site Results From the Eocene Rocks ofAnijima and Nishijima, Bonin Islands

Site			In Situ		Bedding	Corrected			
Name	Rock Type	*N*	Inc.	Dec.	Correction	Inc.	Dec.	a95	k
Site 13	NA	15	29.5	275.3	NA	---	---	3.5	NA
Site 14	dike	6	16.6	78.9	NA	---	---	4.5	NA
Site 14	dike	7	0.4	80.6	NA	---	---	3.6	NA
Site 15	NA	11	6.9	116.4	NA	---	---	21.2	NA
Site 16	NA	29	15.5	344.0	NA	---	---	15.5	NA
Site 17	dike	7	33.7	129.2	NA	---	---	6.6	NA
Site 17	dike	3	18.8	97.0	NA	---	---	4.2	NA
Site 17	pillow lava	9	-6.8	88.9	NA	---	---	4.3	NA
Site 17	pillow lava	8	28.6	83.6	NA	---	---	5.2	NA
Site 17	dike	6	31.5	314.7	NA	---	---	11.7	NA
Site 18	pillow lava	7	-37.4	33.0	NA	---	---	10.8	NA
Site 19	pillow lava	36	-15.1	70.1	NA	---	---	11.8	NA
Site 19	dike	18	9.4	82.5	NA	---	---	15.5	NA
Site 19	dike	6	34.1	94.0	NA	---	---	12.0	NA
Site 19	dike	5	-40.8	101.3	NA	---	---	3.4	NA
Site 19	dike	5	-41.7	100.6	NA	---	---	8.2	NA
Site 19	dike	4	20.3	83.8	NA	---	---	14.3	NA
Site 19	dike	7	43.9	70.1	NA	---	---	6.8	NA
Site 19	dike	9	12.1	67.0	NA	---	---	5.3	NA
Site 20	NA	19	57.0	244.5	NA	---	---	3.0	NA
Site 26*	NA	15	6.3	106.5	NA	---	---	7.9	NA
Site 27*	NA	20	4.4	154.3	NA	---	---	5.5	NA

	N	Inc.	Dec.	a95	k
Mean direction	22	-0.8	91.7	16.9	4.9

Data from Kodama [1981], Kodama et al. [1983], Keating et al. [1983]. Abbreviations are as in Table 1. NA means data not cited or not available.

* Data from Nishijima.

TABLE 13. Site Results From Hahajima, Mukojima, Otootojima, and Yomejima, Bonin Islands

Site Name	Rock Type	N	In Situ Inc.	In Situ Dec.	Bedding Correction	Corrected Inc.	Corrected Dec.	a95	k
				Hahajima					
NK	andesite lava	12	-10.2	210.7	Horizontal	---	---	5.1	77.0
KK	andesite lava	6	22.1	234.9	NA	---	---	13.8	23.6
NS	andesite lava	13	8.4	190.7	Horizontal	---	---	8.9	23.1
MY	andesite lava	11	-2.2	209.7	NA	---	---	5.7	67.1
				Mukojima					
MU-1	pillow lava	8	-25.0	203.7	NA	---	---	9.9	31.8
MU-2	pillow lava	5	14.1	209.2	NA	---	---	2.6	663.1
				Otootojima					
Site 8	NA	13	-7.4	113.1	NA	---	---	6.4	NA
Site 9	dike	6	-83.0	149.5	NA	---	---	15.8	NA
Site 9	pillow lava	11	11.0	95.7	NA	---	---	8.5	NA
Site 11	sandstone	18	-36.2	156.3	NA	---	---	22.8	NA
Site 11	basalt	5	-73.1	77.3	NA	---	---	20.6	NA
Site 12	NA	20	-20.3	46.9	NA	---	---	17.0	NA
				Yomejima					
YM	andesite dike	4	-26.8	232.8	NA	---	---	37.4	7.4

Data from Kodama [1981], Kodama et al. [1983], Keating et al. [1983]. Abbreviations are as in Table 1. NA means data not cited or not available. Mean direction from Hahajima is Inc.=4.7, Dec.=211.1, a95=26.4, and k=13.0.

Yomejima and Mukojima: Yomejima and Mukojima are the northernmost of the Bonin Islands. These islands consists of boninitic and andesitic pillow lavas, hyaloclastites, and dikes similar to the other Bonin islands [*Kodama*, 1981]. Two K-Ar dates of 42 and 22 Ma have been obtained from these islands [*Tsunakawa*, 1983], with the likely age of the rocks being upper Eocene.

Only three sites have been reported from these islands [*Kodama*, 1981], all of which are clockwise deflected between 23° and 52° (Table 13). However, one site (YM) has an $\alpha 95$ of 37° and must be rejected. The small number of sites and lack of structural information precludes any tectonic interpretation of these data.

Paleomagnetic Inclination Studies:

Numerous paleomagnetic studies have been done on DSDP and Ocean Drilling Project (ODP) cores collected from around the Philippine Sea plate [*Louden*, 1977; *Kinoshita*, 1980; *Keating*, 1980; *Keating and Herrero*, 1980; *Bleil*, 1981; *Niitsuma*, 1986; *Haston et al.*, 1989] and from the Izu Peninsula [*Hirooka et al.*, 1985]. Figure 1 shows the locations for the DSDP drill holes from the Philippine Sea plate. The results of these studies are summarized in Table 14.These data suggest that the Philippine Sea plate has undergone as much as 20° of northward translation since the Eocene (Figure 12). The low-latitude position of many of the drill sites and observed paleolatitudes makes it difficult to distinguish between normal and reversely magnetized samples, giving rise to an equatorial ambiguity. That is, it is difficult to determine in which hemisphere the samples were magnetized. However, recent results from ODP cores drilled on the northeast margin of the Philippine Sea plate, are far enough north to eliminate the equatorial ambiguity. They suggest that the plate has undergone at least 15° of northward translation since the Oligocene [*Cisowski et al.*, 1990; *Haston et al.*, 1989]. These inclination studies agree in general with the calculated changes in latitude from the on-land data (Figure 12).

The recent development of an electrical resistivity core orientation tool permits oriented drill cores to be obtained from deep sea drilling. Preliminary results suggest that the northern Bonin arc has undergone approximately 80° of clockwise rotation since the Oligocene [*Cisowski et al.*, 1990].

SUMMARY OF PALEOMAGNETIC RESULTS

All of the reliable paleomagnetic data from the Philippine Sea plate show increasing clockwise declination deflection with time. Various subsets of these data have been used to suggest 60°-70° of clockwise rotation of the West Philippine Sea basin since the middle Oligocene [*Haston et al.*, 1988], continuous clockwise rotation of the Philippine Sea plate since the Eocene [*Keating and Helsley*, 1985], and deformation of the eastern margin related to the collision of buoyant features [*Larson*, 1975; *McCabe and Uyeda*, 1983; *McCabe*, 1984]. We will first summarize the reliable data from the Philippine Sea plate, and then use these data to constrain models for the tectonic evolution of the Philippine Sea plate.

Two reliable mean directions have been obtained from the Palau Islands suggesting 55° of clockwise rotation since the early Miocene and 68° of clockwise rotation since the middle Oligocene. The results from the early Oligocene rocks suggest over 90° of rotation, although these results are less reliable. The strongly rotated Early Oligocene results which imply a short-lived rapid rotation of the Palau islands were interpreted by *Haston et al.* [1988] to represent rotations due to the initial rifting of the Parece-Vela basin. Although the data are compatible with this idea, we feel the data are too few and unreliable to make such an important conclusion.

The data from Guam are generally poor in spite of the substantial effort of several investigators [e.g. *Larson et al.*, 1975; *Keating and Helsley*, 1985]. No reliable results exist from the Plio-Pleistocene Mariana limestone; however; some remagnetized results from the Alutom Formation may suggest a possible small (<20°) late counter-clockwise rotation. The

TABLE 14. Inclination Data From Around the Philippine Sea Plate

Hole	Rock Type	Age m.y.	Observed Paleolatitude	Change in Latitude*	Reference
290	sediments	15.0 ± 2.0	4.2 ± 6.0	-13.4 ± 6.0	*Louden* [1977]
290	sediments	35.0 ± 1.0	26.7 ± 9.7	9.1 ± 9.7	*Louden* [1977]
292	sediments	8.5 ± 1.5	8.8 ± 2.9	-6.8 ± 2.9	*Louden* [1977]
292	sediments	24.0 ± 1.5	7.4 ± 2.1	-8.2 ± 2.1	*Louden* [1977]
292	sediments	28.0 ± 1.0	5.0 ± 2.8	-10.6 ± 2.8	*Louden* [1977]
292	sediments	33.0 ± 1.0	1.7 ± 1.5	-13.9 ± 1.5	*Louden* [1977]
292	sediments	37.5 ± 1.5	0.8 ± 1.2	-14.8 ± 1.2	*Louden* [1977]
292	sediments	41.0 ± 2.0	-4.9 ± 2.2	-20.5 ± 2.2	*Louden* [1977]
294	sediments	24.0 ± 2.0	7.8 ± 8.5	-14.7 ± 8.5	*Louden* [1977]
294	sediments	50.0 ± 5.0	-13.2 ± 6.4	-35.7 ± 6.4	*Louden* [1977]
442-4	both	3.0 ± 2.0	21.0 ± 17.0	-8.0 ± 17.0	*Kinoshita et al.* [1980]
442-4	both	17.0 ± 2.0	18.0 ± 12.0	-11.0 ± 12.0	*Kinoshita et al.* [1980]
445	sediments	2.5 ± 2.5	18.8 ± 12.0	-6.7 ± 12.0	*Kinoshita et al.* [1980]
445	sediments	14.0 ± 9.0	17.4 ± 5.1	-8.1 ± 5.1	*Kinoshita et al.* [1980]
445	sediments	30.0 ± 7.0	10.8 ± 4.4	-14.7 ± 4.4	*Kinoshita et al.* [1980]
445	sediments	44.0 ± 7.0	4.5 ± 2.9	-21.0 ± 2.9	*Kinoshita et al.* [1980]
446	both	44.0 ± 8.0	6.2 ± 4.5	-20.4 ± 4.5	*Kinoshita et al.* [1980]
446A	both	45.0 ± 8.0	4.3 ± 3.0	-20.3 ± 3.0	*Kinoshita et al.* [1980]
447	igneous	33.0 ± 2.0	6.0 ± 13.1	-12.0 ± 13.1	*Keating and Herrero* [1980]
448	sediments	32.5 ± 5.0	4.7 ± 5.4	-11.6 ± 5.4	*Keating* [1980]
448	igneous	33.0 ± 2.0	7.8 ± 10.0	-8.5 ± 10.0	*Keating and Herrero* [1980]
450	sediments	15.0 ± 5.0	7.8 ± 10.8	-10.2 ± 10.8	*Keating* [1980]
453	sediments	3.7 ± 1.0	11.2 ± 1.2	-6.8 ± 1.2	*Bleil* [1981]
454	both	1.2 ± 0.25	12.7 ± 1.7	-5.3 ± 1.7	*Bleil* [1981]
456	both	2.2 ± .25	19.6 ± 3.1	1.6 ± 3.1	*Bleil* [1981]
458	igneous	32.0 ± 2.0	4.5 ± 1.5	-13.5 ± 1.5	*Bleil* [1981]
458	both	32.0 ± 2.0	9.1 ± 0.8	-8.9 ± 0.8	*Bleil* [1981]
459	sediments	17.5 ± 6.5	12.2 ± .75	-5.8 ± .75	*Bleil* [1981]
459	sediments	25.5 ± 4.5	9.2 ± .75	-8.8 ± .75	*Bleil* [1981]
459	igneous	39.0 ± 2.0	6.3 ± 1.2	-11.7 ± 1.2	*Bleil* [1981]
459	igneous	39.0 ± 2.0	0.0 ± 3.0	-18.0 ± 3.0	*Bleil* [1981]
Izu	sediments	4.0 ± .02	34.5 ± 4.4	-0.5 ± 4.4	*Hirooka et al.* [1985]
Izu	limestone	5.6 ± 0.5	25.3 ± 2.0	-9.7 ± 2.0	*Hirooka et al.* [1985]
Izu	sediments	11.0 ± 1.0	21.6 ± 2.4	-13.4 ± 2.4	*Hirooka et al.* [1985]
Izu	sediments	16.0 ± 5.0	21.0 ±12.1	-14.0 ± 12.1	*Hirooka et al.* [1985]
Izu	sediments	16.5 ± 5.0	14.5 ± 4.9	-20.5 ± 4.9	*Hirooka et al.* [1985]

* The difference between the present latitude of the site and the observed paleolatitude.

three results from the Miocene Alifan Limestone are indistinguishable from the reversed present field and are probably remagnetized directions. The best result from Guam comes from the middle Oligocene Alutom Formation, which shows 66° of clockwise deflection. The results from the upper Eocene Facpi Formation exhibit 20°-90° of clockwise deflection, but are rejected from tectonic analysis because of their extreme scatter. A combination of secular variation and lack of structural control is probably the cause of the scatter.

There are three reliable data points from the island of Saipan. Four sites from the Plio-Pleistocene show a slightly counter-clockwise deflected direction; however, it is not distinguishable from the present dipole field direction at the 95% level of confidence. Five results from the Miocene Fina Sisu and Tagpachau Limestone Formations show 28° of clockwise deflection. We interpret this to be a middle Miocene direction. Eleven sites from the upper Eocene Sankakuyama Formation yield a mean direction which is deflected 43° clockwise.

Two of the Bonin Islands yield results which are considered reliable. Twenty six sites from upper Eocene rocks on Chichijima yield a mean direction which is 105° clockwise deflected. Results from 21 sites on Anijima and Nishijima have a mean direction which is 92° clockwise deflected. Four sites from Hahajima have a mean direction which is less clockwise rotated than the other two islands; however, we have not included these results because of the large $\alpha 95$ and small number of sites. Further work is needed to better determine a mean direction for Hahajima and to determine whether the southern portion of the Ogasawara ridge is less clockwise rotated.

DISCUSSION

There are two basic end-member interpretations of the paleomagnetic data from around the Philippine Sea plate: local rotation of small blocks along the plate margin, or rotation of the entire plate. The sharp bend of the Mariana arc has inspired ideas about the role that a collision of a buoyant feature with a subduction zone has on arc shape [*Vogt*, 1973; *Vogt et al.*, 1976]. The early paleomagnetic results from the Mariana and Bonin islands were used to support models for the bending of the island arcs due to collision [*Larson et al.*, 1975; *McCabe and Uyeda*, 1983; *McCabe*, 1984]. *Haston et al.* [1988] used paleomagnetic data from the Palau Islands to suggest that the entire West Philippine Sea province has rotated 60°-70° since the middle Oligocene and questioned whether the rotations along the Mariana and Bonin arcs are due to collision.

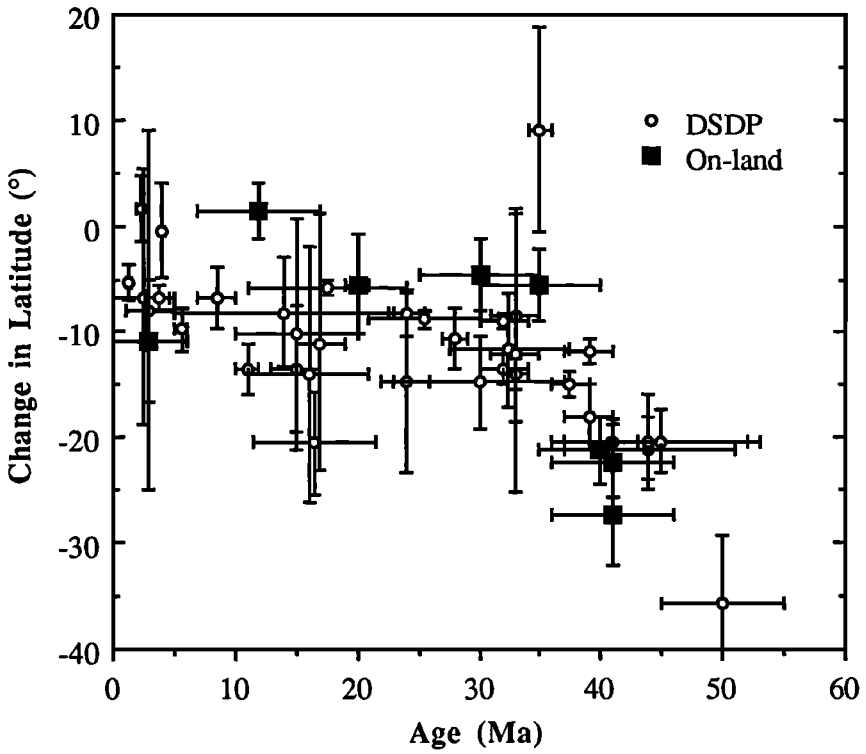


Fig. 12. Data from paleomagnetic inclination data from around the Philippine Sea plate. "Change in Latitude" is defined as the difference between the predicted paleolatitude and the site latitude. This has the effect of normalizing the data independent of the site location.

In an effort to test these two models, we assume that the Palau-Kyushu ridge has remained undeformed since it was left behind by the opening of the Parece-Vela basin at 30 Ma. Although there is evidence which suggests that the Palau-Kyushu Ridge has been offset by 20-30 km in a left-lateral sense at the junction with the Central Basin ridge [*Mrozowski et al.*, 1982], the deformation is undetectable within the resolution of the paleomagnetic method. This implies that the paleomagnetism of Palau records the history of the entire Palau-Kyushu ridge, as well as the West Philippine Sea basin. If no local rotations have occurred along the Bonin and Mariana arcs, then after the Shikoku basin, Parece-Vela basin, and West Mariana trough have been closed and the ridges reconstructed back to their prerifting positions, the paleomagnetic records from the Mariana and Bonin arcs should be identical to Palau. Any differences in the amount of rotation between Palau and the other islands will give an estimate of the amount of local deformation which has occurred since the separation from the Palau-Kyushu ridge.

Reconstruction of the Mariana trough is difficult. *Le Pichon et al.* [1975] determined the pole of rotation for the opening of the basin to be 21.4° N, 138.1° E. *Karig et al.* [1978] presented evidence which showed that the opening of the Mariana trough had a nonrigid component. Using either opening model produces a small counter-clockwise rotation of the Mariana arc around Guam and Saipan. The nonrigid component of opening is probably below the level of resolution for the paleomagnetic method.

There are few constraints on the reconstruction of the West Mariana arc back to the Palau-Kyushu ridge. According to *Mrozowski and Hayes* [1979], spreading occurred between 30 and 17 Ma, with the initiation of spreading occurring in the south and moving northward. The fracture pattern on the basin floor and the south to north initiation of spreading suggest that the pole of rotation for the opening of the basin was to the north; this produces a counter-clockwise (approximately 10°-20°) rotation in the West Mariana and Mariana blocks as the Parece-Vela basin opens.

Despite the lack of a developed back arc basin, there is evidence for large-scale extension between the Ogasawara Ridge and the Bonin arc [*Karig and Moore*, 1975]. The shape of the basin suggests that the Ogasawara Ridge has rotated clockwise relative to the Bonin Arc (Figure 10). The 3000-m isobath was used to reconstruct the two blocks back together.

Spreading in the Shikoku basin rifted the Bonin arc from the Palau-Kyushu Ridge between 25 and 13 Ma, slightly later than the Parece-Vela basin. The fracture and magnetic anomaly patterns of the Shikoku basin [Watts and Weissel, 1975] suggests that the pole of rotation for the basin is far away. Thus very little relative rotation between the Bonin and Ogasawara blocks occurred during the opening of the Shikoku basin.

Extensive deformation has affected the Yap block [*Hawkins and Batiza*, 1977; *Crawford et al.*, 1986], thus making any reconstruction using its present outline inaccurate. For this reason, this portion of the plate margin has been ignored in these reconstructions. This is probably not a serious omission, because no paleomagnetic data come from this block.

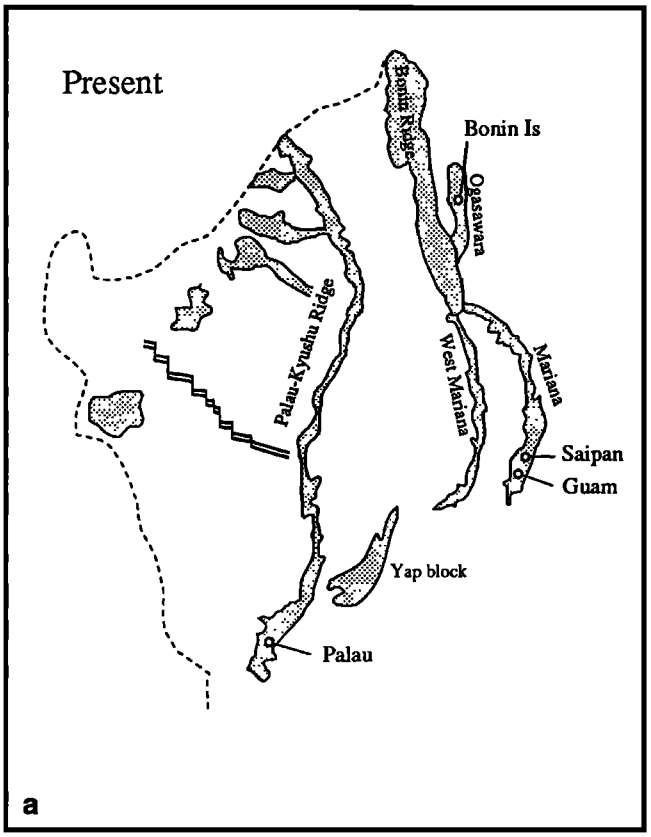
To perform the reconstruction of the blocks back to the Palau-Kyushu ridge, the outlines of the ridges were put back together by visual best fit using two methods, minimizing overlap and minimizing open space (Figure 13). Both geologic and geophysical data from the various back arc basins were used to try and help constrain the reconstructions. Neither method produces a completely satisfactory reconstruction. Table 15 lists the Euler poles used in the two models.

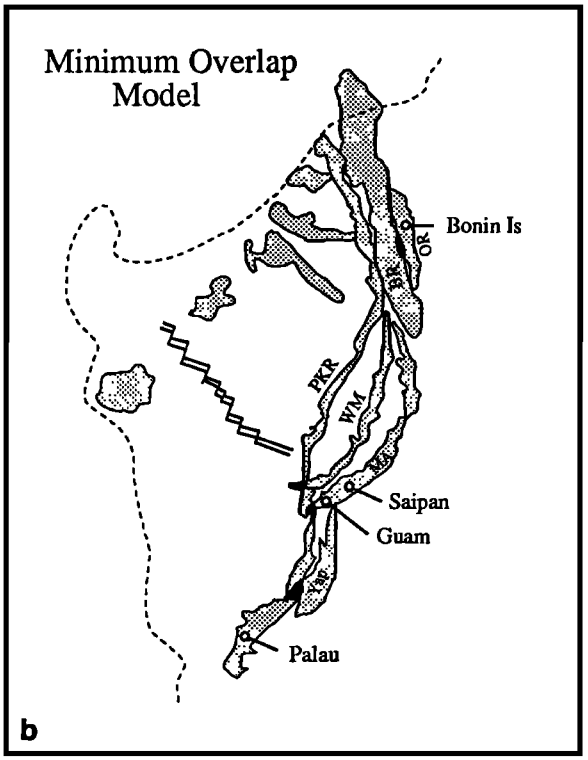
In the minimum overlap case (Figure 13*b*), the Bonin and Ogasawara ridges fit well to the Palau-Kyushu ridge after closure. There is only a slight misfit between the Ogasawara and Bonin ridges, and between the Bonin and Palau-Kyushu

ridges. The fit between the Mariana, West Mariana, and Palau-Kyushu ridges is poor. There is slight overlap of the southern ends of the West Mariana and Mariana ridges over the Palau-Kyushu ridges. However, minimizing the overlap leaves large areas of space, approximately 100-150% of the total width of the three ridges, between the various blocks. The misfit of the Mariana ridge is probably due to a component of nonrigid behavior in this block. *Karig et al.* [1978] suggest that the arc has experienced up to 10% extension along its length; this may explain the overlap on the southern end. The misfit of the West Mariana arc is more serious. There is significant overlap on the southern end, and the ridge intersects the Palau-Kyushu ridge at nearly a right angle. Minimizing the overlap only makes the fit to the north worse.

The minimum space model (Figure 13*c*) uses the same poles as the minimum overlap model for the Bonin and Ogasawara ridges . Although this model does a good job of fitting the main portion of the West Mariana, Mariana, and Palau-Kyushu ridges together, it produces severe overlap in the southern portion of the arcs. If the 10% extensional component [*Karig et al.*, 1978] on the Mariana arc is removed, no overlap exists for this block. However, there is no way to reconcile the overlap in the West Mariana ridge, suggesting that is has undergone significant deformation. The space left between the Mariana, West Mariana, and Palau-Kyushu blocks may be an indication of arc-perpendicular extension prior to the onset of seafloor spreading. We suggest that the minimum space model is the better of the two reconstructions, because this model better accounts for arc-perpendicular extension not recorded in the magnetic anomalies.

To test the local rotation versus plate wide rotation models, the VGPs from around the Philippine Sea plate were compared to each other, in their present in situ positions and after





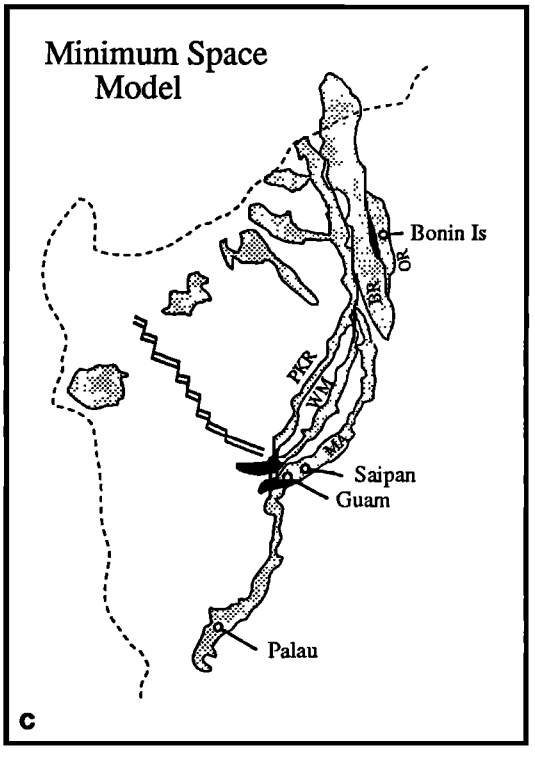


Fig. 13. Models used to reconstruct the East Philippine Sea province. WM is West Mariana Ridge; MA is Mariana ridge; PKR is Palau-Kyushu Ridge; BR is Bonin Ridge; OR is Ogawasara Ridge. A 4000-m isobath was used for the PKR; 3000-m isobath was used for the other blocks. (a) Present position of the major blocks of the Philippine Sea plate. (b) Final position of the blocks after reconstruction of the EPSP by minimizing the overlap in the blocks. Black areas represent areas of overlap. (c) Final position of the blocks after reconstruction of the EPSP by minimizing the space between blocks. Black areas represent areas of overlap.

TABLE 15. Euler Poles Used for EPSP Reconstruction

Block	Fit to:	Visual Best Fit Euler Pole			Age range
		Latitude	Longitude	Angle	
		Minimum Overlap Fit			
Bonin Ridge	Palau-Kyushu Ridge	0.75 N	143.30 E	8.30	24-15
Ogasawara Ridge	Bonin Ridge	24.70 N	141.80 E	8.00	0-3(?)
Mariana Ridge	West Mariana Ridge	21.40 N	138.10 E	-17.00	0-5
West Mariana Ridge	Palau Kyusu Ridge	35.38 S	320.92 E	18.19	30-17
		Minimum Space Fit			
Bonin Ridge	Palau-Kyushu Ridge	0.75 N	143.30 E	8.30	24-15
Ogasawara Ridge	Bonin Ridge	24.70 N	141.80 E	8.00	0-3(?)
Mariana Ridge	West Mariana Ridge	21.40 N	138.10 E	-17.00	0-5
West Mariana Ridge	Palau Kyusu Ridge	32.01 S	321.05 E	24.96	30-17

Relative to a fixed Palau-Kyushu Ridge

closure of the basin by using the minimum space model (Figure 14). The in situ poles show a fair amount of scatter in the older results (Figure 14*a*), however after closure of the basins the poles show better agreement (Figure 14*b*). The paleodirections were calculated from the VGPs and the amount of declination was plotted against time (Figure 15). After closure of the marginal basins, the amount of scatter in the declinations is reduced, which can be seen by comparing Figures 15*a* and 15*b*. The agreement in the amount of declination deflection is good and follows a linear trend. The results are not significantly different when the minimum overlap model is used to perform the reconstruction.

If the deformation model for the bending of the Mariana and Bonin arc is correct, the data from these arcs should not agree with the Palau results when they are reconstructed back to the Palau-Kyushu ridge. Figure 15*b* suggests that although there may have been some deformational rotation of the Mariana and Bonin arcs, the clockwise deflected paleomagnetic results seen around the Philippine Sea plate are related to the rotation of the plate as a whole. Certainly, the collision of the Caroline ridge has cause deformation along the Philippine Sea plate margin, which is evidenced by the collisional features seen on Yap [*Hawkins and Batiza*, 1977; *Crawford et al.*, 1986] and the southern end of the Mariana arc [*Karig et al.*, 1978], but this deformation seems to be a second-order feature compared to the rotation of the plate as a whole. Although simplistic in nature, the minimum space model (Figure 13*c*) suggests that most of the deformation occurred in West Mariana ridge and not the Mariana arc as was suggested by many authors [e.g., *Larson et al.*, 1975; *Karig et al.*, 1978; *McCabe and Uyeda*, 1983, and others]. This may explain the very poor fit of the West Mariana ridge to the Palau-Kyushu ridge. The sharp curvature of the Mariana arc may instead be due to spherical shell buckling, similar to the ideas put forth by *Fukao et al.* [1987], *Yamaoka and Fukao* [1987], *Yamaoka et al.* [1987], and *Yamaoka* [1988], or nonrotational deformation of the arc.

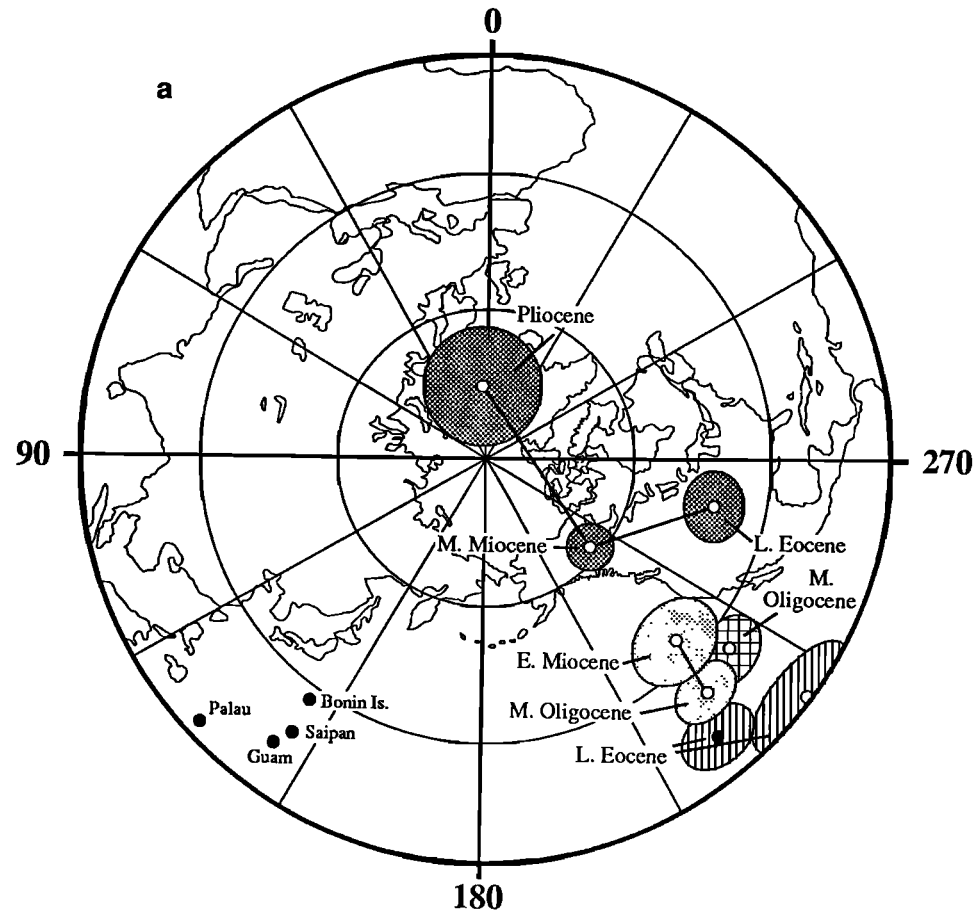
If the Philippine Sea plate has experienced a large amount of rotation and translation, the magnetic anomalies should reflect this motion. There have been many interpretations of the magnetic anomaly data from the West Philippine Sea basin [*Lee and Hilde*, 1971; *Louden*, 1976; *Watts et al.*, 1977; *Bowin et al.*, 1978; *Shih*, 1980*a*; *Mrozowski et al.*, 1982; *Hilde and Lee*, 1984]. However, the complexity of the anomaly pattern and the relatively sparse data coverage have generated a wide range of interpretations (compare *Hilde and Lee* [1984] to *Shih* [1980*a*]]. It is clear that the anomalies in the West Philippine Sea basin are skewed and imply a combination of northward translation and/or rotation. By using the techniques developed by *Schouten and Cande* [1976], an estimate of the paleomagnetic pole can be computed from the anomaly skewness.

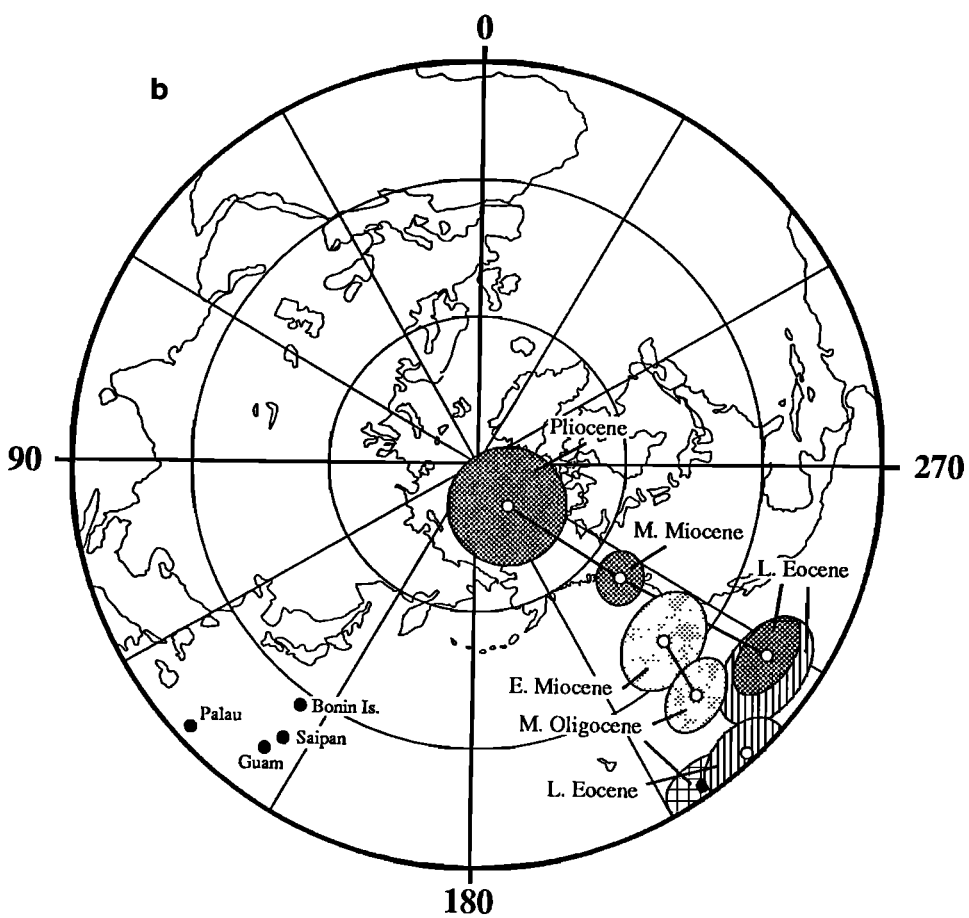
Louden [1977] determined the paleomagnetic pole for the Philippine Sea plate using the anomaly skewness from the West Philippine Sea basin. The pole (Lat.=3°N, Long.=130°W) implies some 90° of clockwise rotation since the Eocene. Shih [1980*a*] also determined a pole (Lat=31° N, Long=106° W) which implies large-scale clockwise rotation (≈60°), but less than Louden's calculated pole. In contrast to these poles, *Bowin et al.* [1978] determined a pole which implies 45° of counter-clockwise rotation, although their data do not exclude the pole determined by Louden. *Watts et al.*'s [1977] skewness analysis suggest less northward motion than either the Shih or Louden models. *Hilde and Lee* [1984] do not consider rotation or the plate but rather suggest a complex history of spreading involving a ridge jump.

Clearly, more magnetic anomaly data are needed from the West Philippine Sea basin to establish a clear history of the spreading. At present it is difficult to assess how the skewness data compare to the on-land paleomagnetic data because of the wide range in interpretations. However, if the Philippine Sea plate has rotated as much as 60° clockwise since the early Miocene, it is difficult to explain the very low amplitude anomalies found in the Parce-Vela basin.

What implications do these data have on the origin and development of the Philippine Sea plate? The data imply that Philippine Sea plate has moved, in general, as a rigid block, and rotated up to 80° clockwise with respect to the dipole since the Eocene. The data do not distinguish between a back arc basin or trapped oceanic crust origin for the West Philippine Sea basin. However, if the trapped ocean crust idea of *Uyeda and Ben-Avraham* [1972] is correct, their model must be modified to account for the rotation of the Philippine Sea plate. In their model, as well as other trapped crust models [i.e., *Uyeda and McCabe*, 1983], a N-S transform fault separating two ridge segments in the Kula-Pacific system changes into a subduction zone when the Pacific plate motion changes in the Eocene.

To be able to incorporate the Philippine Sea plate into global reconstructions, it is necessary to know the pole of rotation for the plate. Two studies have estimated the present pole of rotation for the plate [*Seno*, 1977; *Ranken et al.*, 1984]. Both suggest that the plate is moving westward with respect to Eurasia, and *Ranken et al.* [1984] estimate the plate to be rotating clockwise at a rate of 1.5°-2.0°/yr about a pole near Tokyo. However, it is unclear how far into the past this instantaneous pole can be extrapolated. The paleomagnetic data from the Philippine Sea plate indicate the plate has moved approximately 20° northward since the Eocene and rotated 80° clockwise. Clearly, the present pole of rotation can not be used to describe the motion of the plate in the past, because this would mean that most of the plate would have had more northerly paleolatitudes. The finite pole of rotation must have


a
0
90
270
180
Pliocene
M. Miocene
L. Eocene
M.
Oligocene
E. Miocene
M. Oligocene
L. Eocene
Palau
Bonin Is.
Saipan
Guam


b
0
90
270
180
Pliocene
M. Miocene
L. Eocene
E. Miocene
M. Oligocene
L. Eocene
Palau
Bonin Is.
Saipan
Guam

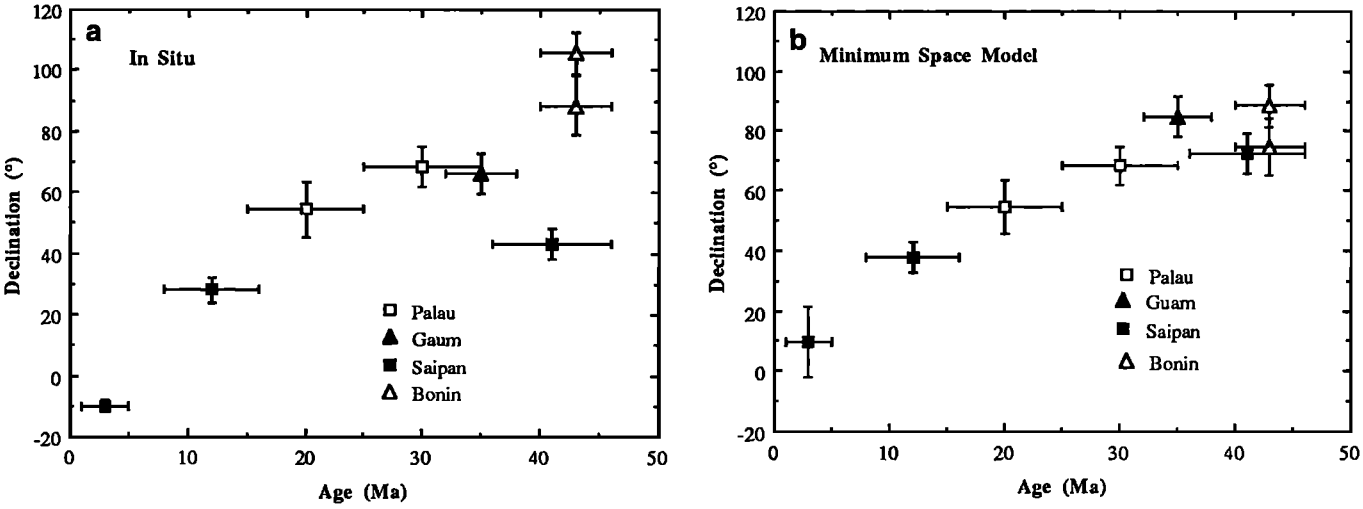


Fig. 15. Declination versus time plot for paleomagnetic data from around the Philippine Sea plate. (a) Pre-reconstructed declinations. (b) Declinations after reconstruction of the EPSP using the minimum space model.

been near the south eastern boundary of the plate; this is the only location that can adequately satisfy the paleomagnetic data. A more rigorous attempt is being made to determine the finite pole of rotation for the Philippine Sea plate with respect to Eurasia.

Clearly, a weakness of these conclusions is the simplistic nature of the models used to reconstruct the EPSP back to the Palau-Kyushu ridge. The key fact is that there is very little difference in the rotational histories seen from the islands of the Philippine Sea plate. We find it unlikely that a complex process such as a collision of a ridge with a subduction zone or nondeformational arc rotation would produce the same rotational history over such a wide area and varying tectonic settings. The collisional deformation model suggests that the islands would have very different rotational histories because they occupy different positions relative to the collision zone. It also seems unlikely that a noncollisional rotation mechanism would produce such similar rotations along such varied locations in the arc. Although deformation may have occurred along the arc, the evidence suggests that the Philippine Sea plate has undergone a large clockwise rotation since the Eocene.

Figures 16*a* (10 Ma) and 16*b* (30 Ma) show two tentative reconstructions for the Philippine Sea plate which incorporate the paleomagnetic data. In the 10-Ma reconstruction, the plate has been back-rotated 30° and moved south 5°. The Mariana trough has been closed. The Marcus-Necker and Caroline ridges have been moved by using the Pacific plate motion [*Engebretson et al.*, 1985]. The Philippines have been largely ignored in these reconstructions, but subduction in the Philippine trench has either just commenced and is propagating northward, or will shortly start up. Figure 16*b* illustrates the geometry of the Philippine Sea plate at 30 Ma. The back arc basins are all closed at this time, and the orientation of the arc is nearly E-W. At first glance this may seem to be impossible, because the Pacific plate (relative to a fixed hot-spot reference frame) is moving westward. However, it is important to compare the relative motion between the Philippine Sea plate and the Pacific plate. Paleolatitude data indicate that the Philippine Sea plate is moving northward at a much faster rate than the Pacific plate, thus there is N-S convergence between the two plates. Therefore subduction in an E-W orientation is not at all unexpected.

We realize that these reconstructions are rather simplistic and have ignored many details in their present state. However, they illustrate the geometry that must have been present if the plate has rotated as a whole. We are currently in the process of doing much more detailed reconstructions which will incorporate data from the Philippines, Taiwan, and Japan, as well as more rigorous reconstructions of the back arc basins.

CONCLUSIONS

The paleomagnetic results from the Philippine Sea plate all show clockwise deflected directions, as well as shallow inclinations. Unfortunately, all of the data come from the margin of the plate. Do these data represent deformation of the plate margin as suggested by *Larson et al.* [1975] and *McCabe and Uyeda* [1983], or do they record the motion of the entire Philippine Sea plate as has been suggested by *Haston et al.* [1988]? By holding the Palau-Kyushu ridge fixed and reconstructing the arcs and arc fragments back to the Palau-Kyushu ridge, the amount of deformational rotation was estimated. All of the results show a similar increase in declination deflection with time, which suggests that the clockwise rotations found around the Philippine Sea plate are not local rotations but rather rotation of the entire Philippine Sea plate.

The data cannot, at this time, determine whether the West Philippine Sea plate formed as a trapped piece of oceanic crust [*Uyeda and Ben-Avraham*, 1972] or as a back arc basin [*Lewis et al.*, 1982; *Seno and Maruyama*, 1984]. However, all of the models require modification to take into account the paleomagnetic data from the Philippine Sea plate. The data require that the Palau-Kyushu ridge originated in a roughly E-W orientation; this is in sharp contrast to the N-S plate boundary invoked by the trapped crust models. The back arc basin

Fig. 14. Equal-area stereographic projection of the VGPs and their associated 67% intervals of confidence from the Philippine Sea plate. Light gray circles are from Palau; dark grey circles are from Saipan, vertically striped circles are from the Bonin Islands; cross-hatched circles are from Guam. (a) Prereconstructed VGPs. (b) VGPs after reconstruction of the EPSP using the minimum space model.

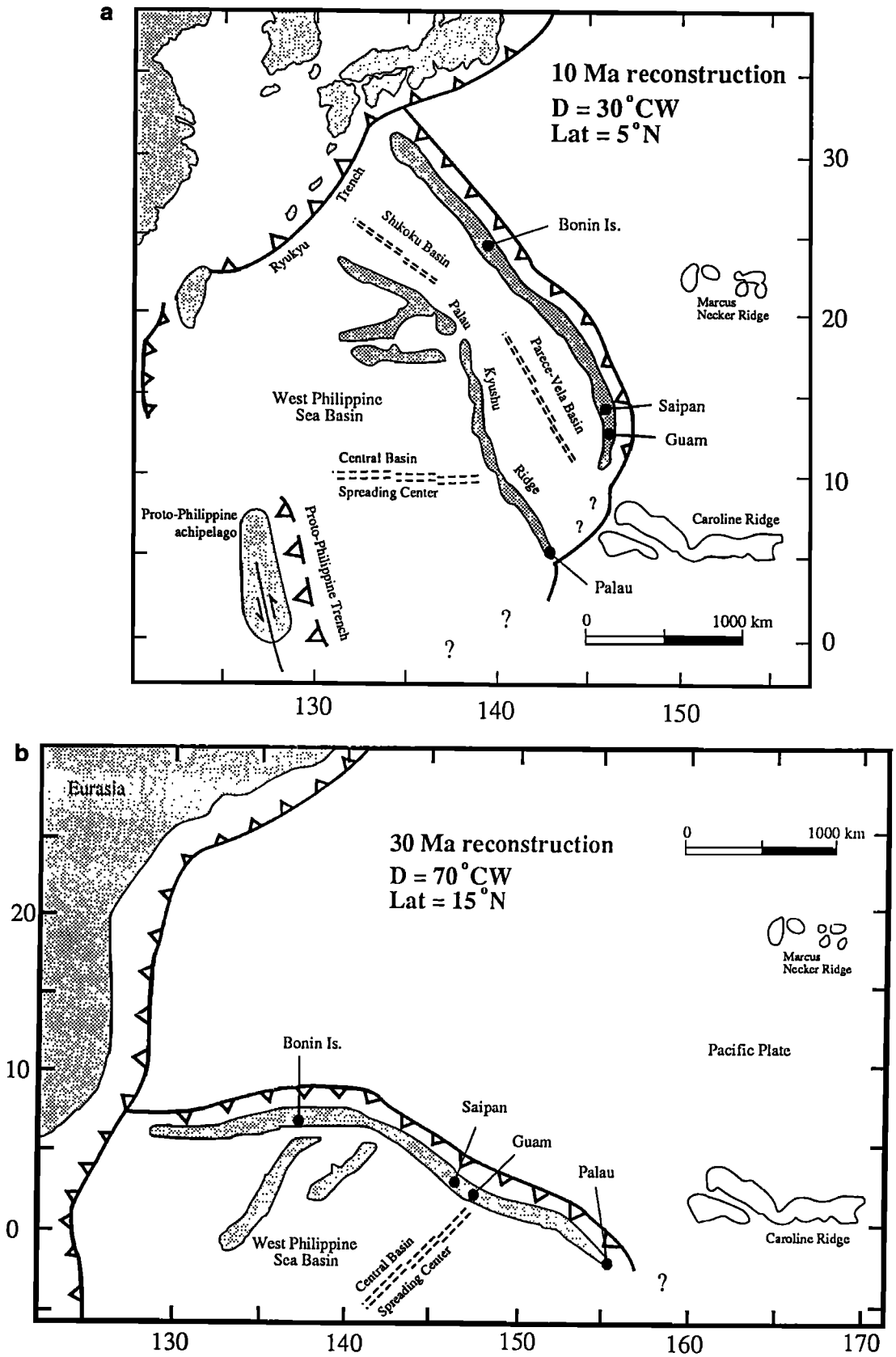


Fig. 16. Tentative reconstructions for the Philippine Sea plate. (a) At 10-Ma reconstruction, 30° of clockwise rotation and 5° of northward translation have been removed; the Mariana trough has been closed. The Marcus-Necker and Caroline Ridges have been moved to their 10-Ma position by using the poles for the Pacific plate from *Engebretson et al.* [1985]. (b) At 30- Ma, 70° of clockwise rotation and 15° of northward translation have been removed; the Shikoku and Parece-Vela basins have been closed. The Marcus-Necker and Caroline Ridges have been moved to their 30-Ma position by using the poles for the Pacific plate from *Engebretson et al.* [1985].

models require the West Philippine Sea basin to have formed behind a west-dipping subduction zone, with the Central Basin ridge oriented generally NEN-SWS.

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